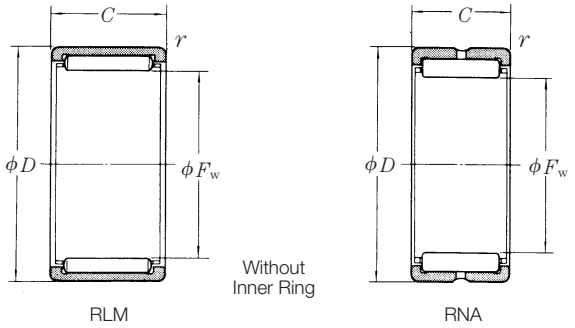
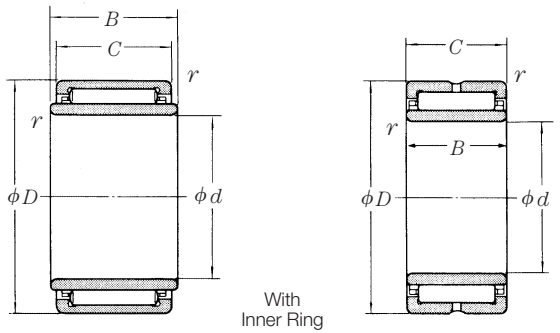
RLM • LM

RNA • NA

Inscribed Circle
Diameter 8 — 15 mm



RLM Without Inner Ring RNA



LM With Inner Ring NA

Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
8	15	10	0.15	4 300	3 800	440	390	60 000
	15	12	0.15	5 500	4 600	560	470	45 000
	15	15	0.3	6 550	5 800	670	590	45 000
	15	16	0.15	7 100	6 350	720	650	45 000
9	16	12	0.3	6 150	5 400	625	550	40 000
	16	16	0.3	7 900	7 450	805	760	40 000
	17	10	0.15	4 700	4 350	470	445	50 000
10	15	10	0.3	4 500	4 900	460	500	36 000
	17	10	0.3	5 350	4 650	545	470	36 000
	17	12	0.15	6 750	6 200	685	635	36 000
	17	15	0.3	8 050	7 800	820	795	36 000
	17	16	0.15	8 650	8 600	885	875	36 000
	19	11	0.2	6 550	6 000	670	615	48 000
12	17	12	0.3	6 150	7 650	625	780	30 000
	19	12	0.3	7 300	7 150	745	730	30 000
	19	15	0.3	8 700	8 950	885	915	30 000
	19	16	0.3	9 400	9 850	955	1 010	30 000
	19	20	0.3	12 000	13 500	1 220	1 380	30 000
	20	11	0.3	7 050	6 850	720	700	38 000
14	22	12	0.3	9 350	9 150	955	935	24 000
	22	13	0.3	9 150	9 950	930	1 010	32 000
	22	16	0.3	12 100	12 700	1 230	1 300	24 000
	22	16	0.3	11 600	13 600	1 190	1 390	32 000
	22	20	0.3	15 500	17 500	1 580	1 790	24 000
	22	22	0.3	16 300	20 900	1 660	2 130	32 000
15	20	8	0.3	4 050	4 800	415	490	24 000
	20	15	0.3	8 100	11 700	825	1 190	24 000
	20	20	0.3	11 100	17 400	1 130	1 770	24 000
	22	12	0.3	8 300	8 900	845	905	24 000
	22	15	0.3	9 900	11 100	1 010	1 140	24 000
	22	20	0.3	13 600	16 800	1 390	1 710	24 000
	23	16	0.3	12 300	14 800	1 250	1 510	24 000
	23	20	0.3	15 500	20 100	1 590	2 050	24 000

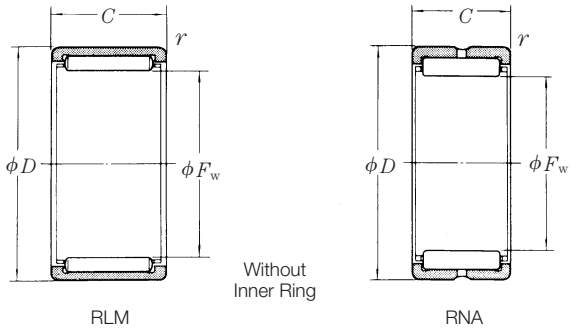
Bearing Numbers					Boundary Dimensions (mm)		Mass (kg) approx	
Without Inner Ring				With Inner Ring				
RLM	RNA49	RNA59	RNA69	LM · NA	d	B	Without Inner Ring	With Inner Ring
	RNA496			—	—	—	0.0070	—
RLM81512-1				**LM81512-1**	5	12	0.0083	0.011
RLM815				—	—	—	0.010	—
RLM81516-1				**LM81516-1**	5	16	0.011	0.015
RLM912				**LM91612-1**	6	12	0.0092	0.013
RLM916				—	—	—	0.011	—
	RNA497			—	—	—	0.0091	—
RLM1010				—	—	—	0.0055	—
RLM101710				—	—	—	0.0082	—
RLM101712-1				**LM101712-1**	7	12	0.0097	0.014
RLM101715				—	—	—	0.012	—
RLM101716-1				**LM101716-1**	7	16	0.013	0.018
	RNA498			**NA498**	8	11	0.012	0.015
RLM1212				**LM1212**	8	12.2	0.0076	0.013
RLM121912				**LM121912**	8	12.2	0.011	0.017
RLM121915				**LM121915**	8	15.2	0.014	0.021
RLM121916-1				**LM121916-1**	9	16	0.014	0.022
RLM121920-1				—	—	—	0.018	—
	RNA499			**NA499**	9	11	0.013	0.017
RLM1412				—	—	—	0.014	—
	RNA4900			**NA4900**	10	13	0.016	0.024
RLM1416				**LM1416**	10	16.2	0.019	0.028
		RNA5900		**NA5900**	10	16	0.022	0.031
RLM1420				**LM1420**	10	20.2	0.024	0.036
			RNA6900	**NA6900**	10	22	0.027	0.040
RLM158				**LM158**	10	8.2	0.0061	0.012
RLM1515				**LM1515**	10	15.2	0.011	0.022
RLM1520				**LM1520**	10	20.2	0.015	0.030
RLM152212				**LM152212**	10	12.2	0.013	0.022
RLM152215				**LM152215**	10	15.2	0.016	0.027
RLM152220				**LM152220**	10	20.2	0.021	0.036
RLM152316-1				—	—	—	0.021	—
RLM152320-1				—	—	—	0.026	—

Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

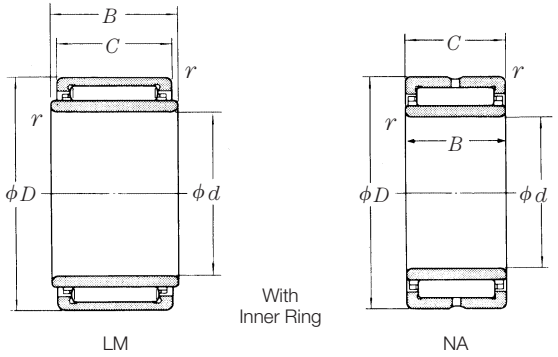
RLM • LM

RNA • NA

Inscribed Circle
Diameter 16 — 21 mm



RLM RNA



LM NA

Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
16	24	13	0.3	10 100	11 700	1 030	1 190	28 000
	24	16	0.3	12 900	14 200	1 310	1 450	22 000
	24	16	0.3	12 800	16 000	1 310	1 630	28 000
	24	20	0.3	16 500	19 500	1 680	1 990	22 000
	24	22	0.3	17 900	24 500	1 830	2 500	28 000
17	22	10	0.3	5 850	7 950	595	810	20 000
	22	20	0.3	11 800	19 500	1 200	1 990	20 000
	24	15	0.5	11 000	13 200	1 120	1 350	20 000
	24	25	0.5	18 200	25 300	1 850	2 580	20 000
	25	16	0.3	13 600	15 500	1 390	1 580	20 000
	25	20	0.3	16 200	21 900	1 650	2 230	20 000
18	25	15	0.5	11 500	14 300	1 170	1 450	20 000
	25	17	0.5	13 300	17 200	1 350	1 750	20 000
	25	20	0.5	15 800	21 500	1 610	2 190	20 000
	25	25	0.5	19 000	27 300	1 940	2 780	20 000
	26	13	0.3	10 500	12 700	1 070	1 290	24 000
	26	16	0.3	12 700	16 200	1 300	1 650	20 000
	26	20	0.3	16 100	22 000	1 640	2 250	20 000
19	27	16	0.5	14 300	17 000	1 460	1 730	18 000
	27	20	0.5	18 300	23 400	1 870	2 380	18 000
20	27	10	0.5	7 950	9 150	810	930	18 000
	27	15	0.5	11 900	15 400	1 220	1 570	18 000
	27	20	0.5	16 400	23 200	1 670	2 370	18 000
	27	20	0.5	16 400	23 200	1 670	2 370	18 000
	27	25	0.5	19 800	29 500	2 010	3 000	18 000
	28	13	0.3	10 800	13 600	1 100	1 390	22 000
	28	16	0.3	14 400	19 700	1 470	2 010	18 000
	28	18	0.3	15 700	21 900	1 600	2 240	22 000
	28	20	0.5	18 200	23 500	1 860	2 400	18 000
	28	23	0.3	19 300	28 600	1 960	2 920	22 000
21	29	16	0.5	14 900	18 500	1 520	1 880	17 000
	29	20	0.5	19 100	25 400	1 950	2 590	17 000

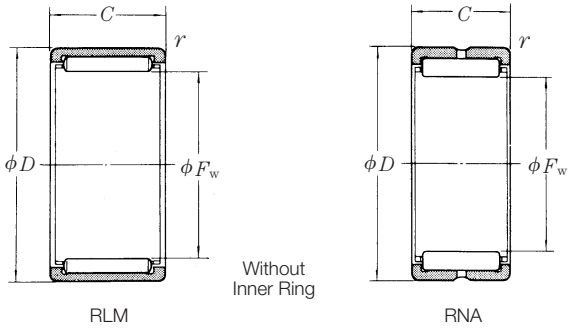
Bearing Numbers					Boundary Dimensions (mm)		Mass (kg) approx	
Without Inner Ring				With Inner Ring				
RLM	RNA49	RNA59	RNA69	LM · NA	d	B	Without Inner Ring	With Inner Ring
	RNA4901			**NA4901**	12	13	0.018	0.027
RLM1616				**LM1616**	12	16.2	0.021	0.032
		RNA5901		**NA5901**	12	16	0.024	0.035
RLM1620				**LM1620**	12	20.2	0.027	0.041
			RNA6901	**NA6901**	12	22	0.030	0.045
RLM1710				**LM1710**	12	10.2	0.0080	0.017
RLM1720				**LM1720**	12	20.2	0.016	0.034
RLM172415				**LM172415**	12	15.2	0.018	0.032
RLM172425				**LM172425**	12	25.2	0.030	0.052
RLM172516-1				—	—	—	0.023	—
RLM172520-1				—	—	—	0.029	—
RLM1815				**LM1815**	15	15.2	0.019	0.028
RLM1817				**LM1817**	15	17.2	0.021	0.031
RLM1820				**LM1820**	15	20.2	0.025	0.037
RLM1825				**LM1825**	15	25.2	0.032	0.047
	RNA49/14			—	—	—	0.020	—
RLM182616-1				—	—	—	0.024	—
RLM182620-1				—	—	—	0.030	—
RLM1916				**LM1916**	15	16.2	0.025	0.039
RLM1920				**LM1920**	15	20.2	0.031	0.048
RLM2010				**LM2010**	15	10.2	0.014	0.025
RLM2015				**LM2015**	15	15.2	0.021	0.037
RLM2020				**LM2020**	15	20.2	0.028	0.049
RLM2020				**LM2020-1**	15	20	0.028	0.048
RLM2025				**LM2025**	15	25.2	0.035	0.061
	RNA4902			**NA4902**	15	13	0.021	0.035
RLM202816-1				—	—	—	0.026	—
		RNA5902		**NA5902**	15	18	0.032	0.051
RLM202820				**LM202820**	15	20.2	0.033	0.055
			RNA6902	**NA6902**	15	23	0.039	0.064
RLM2116				**LM2116**	17	16.2	0.027	0.042
RLM2120				**LM2120**	17	20.2	0.034	0.053

Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

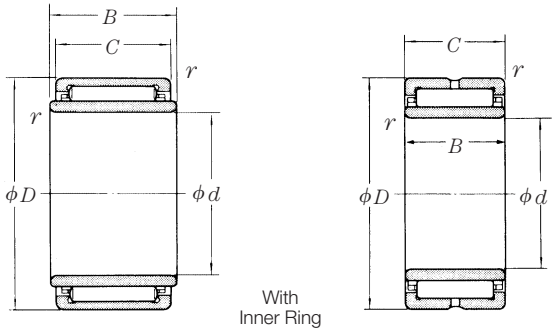
RLM • LM

RNA • NA

Inscribed Circle
Diameter 22 — 29 mm



RLM Without Inner Ring RNA



LM With Inner Ring NA

Boundary Dimensions (mm)				Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min⁻¹)
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
22	29	15	0.5	12 900	17 500	1 310	1 780	16 000
	29	20	0.5	17 700	26 400	1 810	2 690	16 000
	29	25	0.5	21 300	33 500	2 170	3 400	16 000
	30	13	0.3	11 600	15 400	1 190	1 570	20 000
	30	16	0.5	15 600	19 800	1 590	2 020	16 000
	30	18	0.3	16 800	24 800	1 720	2 530	20 000
	30	20	0.5	20 000	27 200	2 030	2 780	16 000
	30	23	0.3	20 700	32 500	2 110	3 300	20 000
24	31	20	0.5	18 200	28 100	1 860	2 870	15 000
	31	28	0.5	23 700	39 500	2 420	4 000	15 000
	32	16	0.5	16 100	21 300	1 650	2 170	15 000
	32	20	0.5	20 700	29 200	2 110	2 980	15 000
25	32	12	0.5	10 300	13 700	1 050	1 400	14 000
	32	20	0.5	18 800	29 700	1 920	3 050	14 000
	32	25	0.5	22 700	37 500	2 310	3 850	14 000
	33	16	0.5	16 800	22 600	1 710	2 300	14 000
	33	20	0.5	21 500	31 000	2 190	3 150	14 000
	33	25	0.5	25 900	39 500	2 640	4 050	14 000
	37	17	0.3	19 700	22 900	2 010	2 340	18 000
	37	23	0.3	27 800	35 500	2 830	3 650	18 000
	37	30	0.3	36 500	50 500	3 700	5 150	18 000
26	34	20	0.5	21 400	31 000	2 180	3 200	13 000
28	35	20	0.5	19 900	33 000	2 030	3 350	12 000
	35	25	0.3	23 900	42 000	2 440	4 250	12 000
	37	20	0.5	24 200	33 500	2 460	3 450	12 000
	37	30	0.5	34 000	52 500	3 450	5 350	12 000
	39	17	0.3	22 400	30 500	2 290	3 150	15 000
	39	23	0.3	28 300	41 500	2 890	4 200	15 000
	39	30	0.3	37 000	58 500	3 800	6 000	15 000
29	38	20	0.5	25 100	36 000	2 560	3 650	12 000
	38	20	0.5	23 400	36 500	2 390	3 700	12 000
	38	30	0.5	35 500	55 500	3 600	5 650	12 000
	38	30	0.3	32 000	54 000	3 250	5 550	12 000

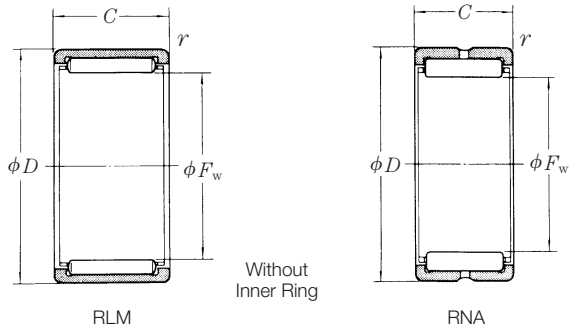
Bearing Numbers					Boundary Dimensions (mm)		Mass (kg) approx	
Without Inner Ring				With Inner Ring				
RLM	RNA49	RNA59	RNA69	LM · NA	d	B	Without Inner Ring	With Inner Ring
RLM2215				**LM2215**	17	15.2	0.023	0.041
RLM2220				**LM2220**	17	20.2	0.030	0.054
RLM2225				**LM2225**	17	25.2	0.038	0.068
	RNA4903			**NA4903**	17	13	0.023	0.038
RLM223016				**LM223016**	17	16.2	0.028	0.045
		RNA5903		**NA5903**	17	18	0.034	0.055
RLM223020				**LM223020**	17	20.2	0.035	0.060
			RNA6903	**NA6903**	17	23	0.041	0.068
RLM2420				**LM2420**	20	20.2	0.033	0.054
RLM2428				**LM2428**	20	28.2	0.046	0.076
RLM243216				**LM243216**	20	16.2	0.030	0.048
RLM243220				**LM243220**	20	20.2	0.038	0.060
RLM2512				**LM2512**	20	12.2	0.020	0.036
RLM2520				**LM2520**	20	20.2	0.034	0.061
RLM2525				**LM2525**	20	25.2	0.042	0.076
RLM253316-1				—	—	—	0.032	—
RLM253320				**LM253320**	20	20.2	0.040	0.068
RLM253325				**LM253325**	20	25.2	0.050	0.085
	RNA4904			**NA4904**	20	17	0.055	0.077
		RNA5904		**NA5904**	20	23	0.089	0.12
			RNA6904	**NA6904**	20	30	0.098	0.14
RLM2620				**LM2620**	22	16.2	0.041	0.065
RLM2820				**LM2820**	22	20.2	0.038	0.062
RLM2825				**LM2825**	22	25.2	0.047	0.092
RLM283720				**LM283720**	22	20.2	0.050	0.087
RLM283730				**LM283730**	22	30.2	0.075	0.13
	RNA49/22			**NA49/22**	22	17	0.056	0.086
		RNA59/22		**NA59/22**	22	23	0.091	0.135
			RNA69/22	**NA69/22**	22	30	0.096	0.15
RLM2920				**LM2920**	25	20.2	0.052	0.079
RLM293820-1				**LM293820-1**	25	20	0.052	0.078
RLM2930				**LM2930**	25	30.2	0.078	0.118
RLM293830				**LM293830**	25	30.2	0.078	0.117

Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

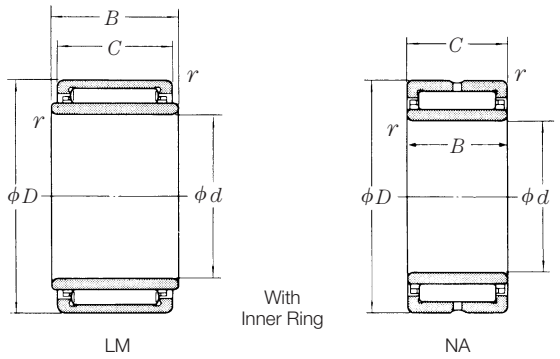
RLM • LM

RNA • NA

Inscribed Circle
Diameter 30 — 38 mm



RLM RNA



LM NA

Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
30	37	15	0.5	14 800	23 000	1 500	2 340	12 000
	37	20	0.5	20 300	34 500	2 070	3 550	12 000
	37	25	0.5	24 500	44 000	2 490	4 500	12 000
	40	20	0.5	25 000	36 000	2 550	3 650	12 000
	40	25	0.5	30 000	46 000	3 100	4 650	12 000
	40	30	0.5	35 000	56 000	3 600	5 700	12 000
	42	17	0.3	21 400	26 800	2 180	2 740	14 000
	42	23	0.3	30 000	41 500	3 100	4 250	14 000
	42	30	0.3	39 500	59 000	4 050	6 050	14 000
32	42	20	0.5	25 800	38 000	2 630	3 900	11 000
	42	30	0.5	36 500	59 000	3 700	6 050	11 000
	45	17	0.3	22 200	28 700	2 270	2 930	13 000
	45	23	0.3	31 500	44 500	3 200	4 550	13 000
	45	30	0.3	41 000	63 500	4 200	6 450	13 000
35	42	20	0.5	22 300	41 000	2 270	4 200	10 000
	42	30	0.5	31 000	63 500	3 200	6 450	10 000
	45	20	0.5	27 500	42 500	2 800	4 350	10 000
	45	25	0.5	33 000	54 500	3 400	5 550	10 000
	45	30	0.5	38 500	66 000	3 950	6 750	10 000
	47	17	0.3	23 900	32 500	2 430	3 300	12 000
	47	23	0.3	33 500	50 500	3 450	5 150	12 000
	47	30	0.3	44 000	71 500	4 500	7 300	12 000
37	47	20	0.6	28 200	45 000	2 880	4 550	9 500
	47	30	0.6	39 500	69 500	4 050	7 100	9 500
	47	30	0.6	39 500	69 500	4 050	7 100	9 500
38	48	15	0.6	20 900	30 500	2 130	3 150	9 000
	48	20	0.6	29 000	47 000	2 960	4 800	9 000
	48	25	0.6	35 000	60 000	3 600	6 100	9 000
	48	30	0.6	41 000	73 000	4 150	7 450	9 000

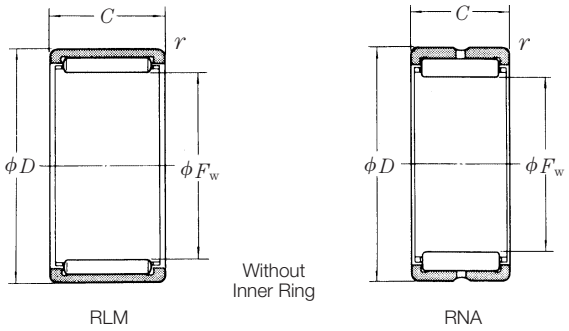
Bearing Numbers					Boundary Dimensions (mm)		Mass (kg) approx	
Without Inner Ring				With Inner Ring				
RLM	RNA49	RNA59	RNA69	LM · NA	d	B	Without Inner Ring	With Inner Ring
RLM3015				**LM3015**	25	15.2	0.030	0.055
RLM3020				**LM3020**	25	20.2	0.040	0.073
RLM3025				**LM3025**	25	25.2	0.050	0.092
RLM304020				**LM304020**	25	20.2	0.060	0.093
RLM304025				**LM304025**	25	25.2	0.075	0.12
RLM304030				**LM304030**	25	30.2	0.090	0.14
	RNA4905			**NA4905**	25	17	0.063	0.091
		RNA5905		**NA5905**	25	23	0.10	0.14
			RNA6905	**NA6905**	25	30	0.11	0.16
RLM3220				**LM3220**	28	20.2	0.064	0.090
RLM3230				**LM3230**	28	30.2	0.096	0.14
	RNA49/28			**NA49/28**	28	17	0.076	0.099
		RNA59/28		**NA59/28**	28	23	0.11	0.145
			RNA69/28	**NA69/28**	28	30	0.13	0.175
RLM3520				**LM3520**	30	20.2	0.046	0.085
RLM3530				**LM3530**	30	30.2	0.070	0.13
RLM354520				**LM354520**	30	20.2	0.069	0.11
RLM354525				**LM354525**	30	25.2	0.086	0.135
RLM354530				**LM354530**	30	30.2	0.10	0.16
	RNA4906			**NA4906**	30	17	0.072	0.105
		RNA5906		**NA5906**	30	23	0.11	0.15
			RNA6906	**NA6906**	30	30	0.13	0.19
RLM3720				**LM3720**	32	20.3	0.072	0.115
RLM3730				**LM3730**	32	30.3	0.11	0.175
RLM374730-1				**LM374730-1**	32	30	0.11	0.17
RLM3815				**LM3815**	32	15.3	0.056	0.094
RLM3820				**LM3820**	32	20.3	0.074	0.125
RLM3825				**LM3825**	32	25.3	0.093	0.16
RLM3830				**LM3830**	32	30.3	0.11	0.195

Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

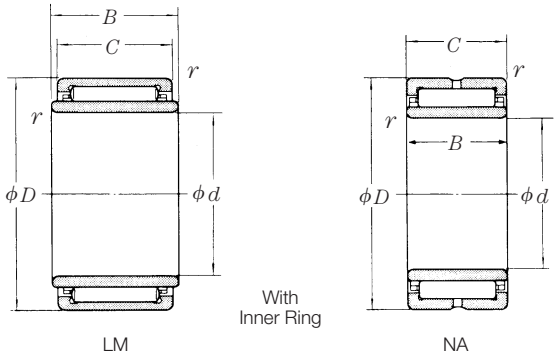
RLM • LM

RNA • NA

Inscribed Circle
Diameter 40 — 50 mm



RLM RNA



LM NA

Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
40	50	15	0.6	21 400	32 000	2 180	3 300	9 000
	50	20	0.6	29 700	49 000	3 050	5 000	9 000
	50	20	0.6	29 700	49 000	3 050	5 000	9 000
	50	25	0.6	36 000	62 500	3 650	6 400	9 000
	50	30	0.6	42 000	76 500	4 250	7 800	9 000
	52	20	0.6	29 900	45 000	3 050	4 600	10 000
	52	27	0.6	40 500	66 000	4 100	6 750	10 000
	52	36	0.6	56 000	101 000	5 700	10 300	10 000
42	52	20	0.3	32 000	54 500	3 250	5 600	8 500
	52	30	0.3	44 500	84 500	4 550	8 600	8 500
	55	20	0.6	30 500	47 500	3 100	4 800	10 000
	55	27	0.6	41 500	69 500	4 200	7 100	10 000
	55	36	0.6	57 500	106 000	5 850	10 900	10 000
43	53	20	0.3	32 000	55 000	3 250	5 600	8 000
	53	30	0.3	44 500	84 500	4 550	8 600	8 000
45	55	20	0.6	31 000	53 500	3 150	5 500	8 000
	55	25	0.6	37 500	68 500	3 800	7 000	8 000
	55	30	0.6	43 500	83 500	4 450	8 500	8 000
	58	20	0.6	34 000	56 000	3 500	5 700	9 500
47	57	20	0.3	33 000	59 500	3 350	6 050	7 500
	57	30	0.3	46 500	91 500	4 700	9 350	7 500
48	62	22	0.6	39 000	61 500	3 950	6 300	9 000
	62	30	0.6	54 500	95 000	5 550	9 700	9 000
	62	40	0.6	72 000	137 000	7 350	13 900	9 000
50	60	20	0.6	33 000	60 500	3 350	6 150	7 100
	60	30	0.6	46 500	94 000	4 750	9 550	7 100
	62	20	0.6	35 500	60 500	3 600	6 150	7 100
	62	25	0.6	43 000	77 500	4 400	7 900	7 100
	62	25	0.6	43 000	77 500	4 400	7 900	7 100
	62	30	0.6	50 000	94 500	5 100	9 600	7 100
	62	35	0.6	66 500	136 000	6 800	13 900	7 100
	65	22	0.6	43 000	72 000	4 400	7 300	8 500

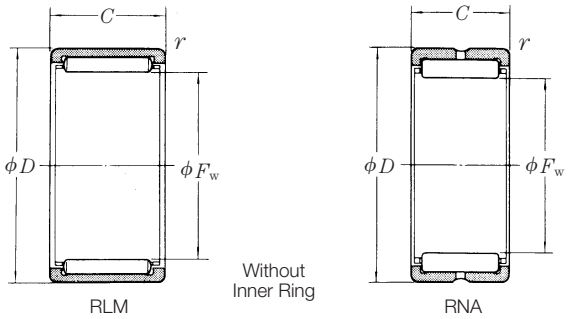
Bearing Numbers					Boundary Dimensions (mm)		Mass (kg) approx	
Without Inner Ring				With Inner Ring				
RLM	RNA49	RNA59	RNA69	LM · NA	d	B	Without Inner Ring	With Inner Ring
RLM4015				**LM4015**	35	15.3	0.058	0.092
RLM4020				**LM4020**	35	20.3	0.078	0.125
RLM405020-1				**LM405020-1**	35	20	0.125	0.125
RLM4025				**LM4025**	35	25.3	0.097	0.155
RLM4030				**LM4030**	35	30.3	0.12	0.19
	RNA49/32			**NA49/32**	32	20	0.092	0.16
		RNA59/32		**NA59/32**	32	27	0.15	0.24
			RNA69/32	**NA69/32**	32	36	0.17	0.29
RLM425220-1				—	—	—	0.081	—
RLM425230-1				—	—	—	0.12	—
	RNA4907			**NA4907**	35	20	0.11	0.17
		RNA5907		**NA5907**	35	27	0.175	0.25
			RNA6907	**NA6907**	35	36	0.20	0.315
RLM435320-1				**LM435320-1**	38	20	0.082	0.132
RLM435330-1				**LM435330-1**	38	30	0.125	0.199
RLM4520				**LM4520**	40	20.3	0.086	0.14
RLM4525				**LM4525**	40	25.3	0.11	0.17
RLM4530				**LM4530**	40	30.3	0.13	0.21
	RNA49/38			—	—	—	0.12	—
RLM475720-1				**LM475720-1**	42	20	0.089	0.12
RLM475730-1				**LM475730-1**	42	30	0.14	0.22
	RNA4908			**NA4908**	40	22	0.15	0.24
		RNA5908		**NA5908**	40	30	0.23	0.355
			RNA6908	**NA6908**	40	40	0.265	0.435
RLM5020				**LM5020**	42	20.3	0.098	0.19
RLM5030				**LM5030**	42	30.3	0.15	0.28
RLM506220				**LM506220**	42	20.3	0.12	0.21
RLM506225				**LM506225**	42	25.3	0.155	0.265
RLM506225				**LM506225-1**	45	25	0.155	0.22
RLM506230				**LM506230**	42	30.3	0.18	0.315
RLM506235-1				**LM506235-1**	45	35	0.21	0.31
	RNA49/42			—	—	—	0.17	—

Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

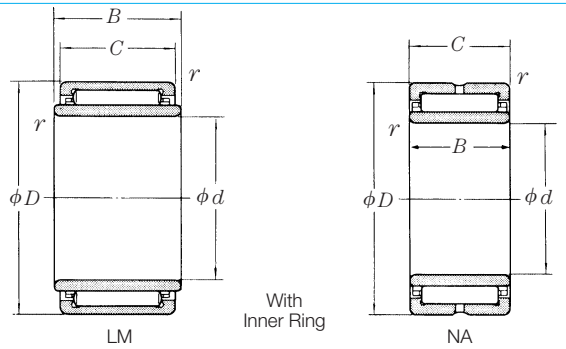
RLM • LM

RNA • NA

Inscribed Circle
Diameter 52 — 73 mm



RLM — Without Inner Ring — RNA



LM — With Inner Ring — NA

Boundary Dimensions (mm)				Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min⁻¹)
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
52	68	22	0.6	41 000	67 500	4 150	6 900	8 000
	68	30	0.6	57 000	104 000	5 800	10 600	8 000
	68	40	0.6	76 000	149 000	7 750	15 200	8 000
55	65	30	0.6	49 000	104 000	5 000	10 600	6 300
	65	40	0.6	64 000	146 000	6 550	14 900	6 300
	67	20	0.6	38 000	68 000	3 850	6 900	6 300
	67	25	0.6	46 000	87 000	4 700	8 850	6 300
	68	25	0.5	48 000	92 500	4 900	9 450	6 300
	68	35	0.3	63 500	132 000	6 500	13 500	6 300
	70	22	0.6	45 000	78 000	4 600	7 950	7 500
58	72	22	0.6	42 500	73 500	4 350	7 500	7 100
	72	30	0.6	59 500	113 000	6 050	11 500	7 100
	72	40	0.6	79 000	163 000	8 050	16 600	7 100
60	70	40	0.6	66 000	156 000	6 750	15 900	6 000
	72	25	0.3	50 000	99 500	5 100	10 100	6 000
	72	30	0.6	58 000	120 000	5 900	12 300	6 000
	72	35	0.3	65 500	142 000	6 700	14 400	6 000
	72	40	0.6	73 000	162 000	7 450	16 600	6 000
	75	22	0.6	48 000	87 500	4 900	8 900	7 100
63	80	25	1	53 500	87 500	5 450	8 950	6 700
	80	34	1	73 500	133 000	7 500	13 600	6 700
	80	45	1	93 500	181 000	9 550	18 500	6 700
65	78	35	0.6	67 500	151 000	6 900	15 400	5 300
	82	25	1	60 500	105 000	6 200	10 700	6 300
68	82	25	0.6	56 500	112 000	5 750	11 500	5 000
	82	35	0.6	78 000	169 000	7 950	17 300	5 000
	85	25	1	56 000	95 500	5 700	9 750	6 300
	85	34	1	77 500	145 000	7 900	14 800	6 300
	85	45	1	98 000	197 000	10 000	20 100	6 300
70	88	25	1	63 000	113 000	6 450	11 500	6 000
72	90	25	1	58 500	103 000	5 950	10 500	5 600
	90	34	1	81 000	157 000	8 250	16 000	5 600
	90	45	1	103 000	213 000	10 500	21 800	5 600
73	90	25	1	64 500	117 000	6 550	12 000	4 800
	90	35	1	88 500	177 000	9 000	18 000	4 800

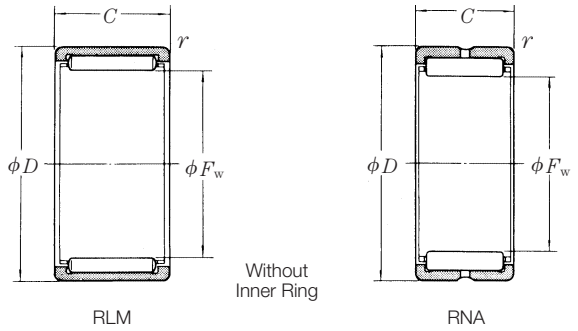
Bearing Numbers					Boundary Dimensions (mm)		Mass (kg) approx	
Without Inner Ring				With Inner Ring				
RLM	RNA49	RNA59	RNA69	LM · NA	d	B	Without Inner Ring	With Inner Ring
	RNA4909			**NA4909**	45	22	0.19	0.28
		RNA5909		**NA5909**	45	30	0.27	0.39
			RNA6909	**NA6909**	45	40	0.335	0.495
RLM5530				**LM5530**	45	30.3	0.16	0.34
RLM5540				**LM5540**	45	40.3	0.215	0.46
RLM556720				**LM556720**	45	20.3	0.13	0.25
RLM556725				**LM556725**	45	25.3	0.165	0.32
RLM556825-1				**LM556825-1**	50	25	0.18	0.27
RLM556835-1				**LM556835-1**	50	35	0.25	0.37
	RNA49/48			—	—	—	0.18	—
	RNA4910			**NA4910**	50	22	0.18	0.295
		RNA5910		**NA5910**	50	30	0.25	0.405
			RNA6910	**NA6910**	50	40	0.32	0.53
RLM6040				**LM6040**	50	40.3	0.235	0.505
RLM607225				**LM607225-1**	55	25	0.175	0.26
RLM607230				**LM607230**	50	30.3	0.21	0.41
RLM607235-1				**LM607235-1**	55	35	0.245	0.37
RLM607240				**LM607240**	50	40.3	0.28	0.545
	RNA49/52			—	—	—	0.20	—
	RNA4911			**NA4911**	55	25	0.26	0.40
		RNA5911		**NA5911**	55	34	0.37	0.56
			RNA6911	**NA6911**	55	45	0.475	0.73
RLM657835-1				—	—	—	0.29	—
	RNA49/58			—	—	—	0.27	—
RLM688225-1				**LM688225-1**	60	25	0.23	0.39
RLM688235-1				**LM688235-1**	60	35	0.325	0.54
	RNA4912			**NA4912**	60	25	0.28	0.435
		RNA5912		**NA5912**	60	34	0.415	0.625
			RNA6912	**NA6912**	60	45	0.485	0.76
	RNA49/62			—	—	—	0.31	—
	RNA4913			**NA4913**	65	25	0.32	0.465
		RNA5913		**NA5913**	65	34	0.48	0.675
			RNA6913	**NA6913**	65	45	0.53	0.79
RLM739025-1				—	—	—	0.305	—
RLM739035-1				**LM739035-1**	65	35	0.43	0.67

Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

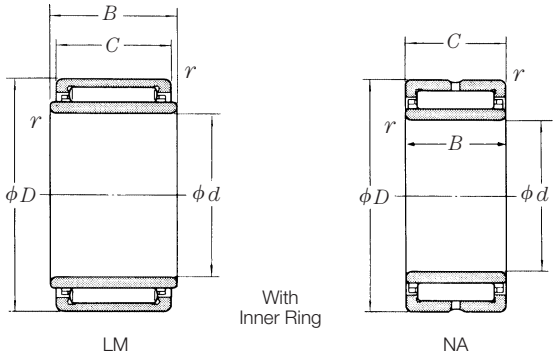
RLM • LM

RNA • NA

Inscribed Circle
Diameter 75 — 115 mm



RLM RNA



LM NA

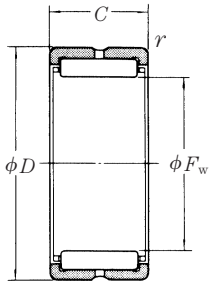
Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
75	92	25	1	64 000	118 000	6 550	12 000	4 800
	92	35	1	88 000	177 000	8 950	18 100	4 800
	95	30	1	83 500	148 000	8 550	15 100	5 600
80	95	25	1	62 500	125 000	6 400	12 800	4 500
	95	35	1	86 000	189 000	8 800	19 200	4 500
	100	30	1	80 500	143 000	8 200	14 600	5 300
	100	40	1	107 000	206 000	10 900	21 000	5 300
	100	54	1	143 000	298 000	14 500	30 500	5 300
85	105	25	1	79 000	143 000	8 100	14 600	4 000
	105	30	1	84 000	155 000	8 600	15 800	5 000
	105	35	1	109 000	215 000	11 100	21 900	4 000
	105	40	1	112 000	222 000	11 400	22 700	5 000
	105	54	1	149 000	325 000	15 200	33 000	5 000
90	110	25	1	82 500	154 000	8 400	15 700	4 000
	110	30	1	87 500	166 000	8 950	17 000	4 500
	110	40	1	116 000	239 000	11 900	24 400	4 500
	110	54	1	155 000	345 000	15 900	35 500	4 500
95	115	26	1	85 500	164 000	8 700	16 700	3 600
	115	30	1	95 500	189 000	9 750	19 300	4 300
100	120	26	1	86 000	168 000	8 750	17 100	3 600
	120	35	1.1	104 000	214 000	10 600	21 800	4 000
	120	46	1.1	138 000	310 000	14 100	31 500	4 000
	120	63	1.1	174 000	415 000	17 800	42 500	4 000
105	125	35	1.1	108 000	228 000	11 000	23 300	4 000
	125	46	1.1	143 000	330 000	14 600	33 500	4 000
	125	63	1.1	181 000	445 000	18 400	45 000	4 000
110	130	30	1	101 000	213 000	10 300	21 800	3 200
	130	35	1.1	111 000	242 000	11 400	24 700	3 800
	130	40	1	134 000	305 000	13 700	31 500	3 200
	130	46	1.1	148 000	350 000	15 100	35 500	3 800
	130	63	1.1	187 000	470 000	19 100	48 000	3 800
115	140	40	1.1	144 000	295 000	14 700	30 000	3 600
	140	54	1.1	193 000	430 000	19 700	43 500	3 600

Bearing Numbers					Boundary Dimensions (mm)		Mass (kg) approx	
	Without Inner Ring			With Inner Ring				
RLM	RNA49	RNA59	RNA69	LM · NA	d	B	Without Inner Ring	With Inner Ring
RLM759225-1				—	—	—	0.315	—
RLM759235-1				—	—	—	0.44	—
	RNA49/68			—	—	—	0.46	—
RLM809525-1				**LM809525-1**	70	25	0.29	0.52
RLM809535-1				**LM809535-1**	70	35	0.405	0.73
	RNA4914			**NA4914**	70	30	0.47	0.74
		RNA5914		**NA5914**	70	40	0.69	1.05
			RNA6914	**NA6914**	70	54	0.89	1.4
RLM8510525-1				**LM8510525-1**	75	25	0.42	0.67
	RNA4915			**NA4915**	75	30	0.50	0.79
RLM8510535-1				**LM8510535-1**	75	35	0.59	0.93
		RNA5915		**NA5915**	75	40	0.735	1.1
			RNA6915	**NA6915**	75	54	0.96	1.5
RLM9011025-1				**LM9011025-1**	80	25	0.44	0.70
	RNA4916			**NA4916**	80	30	0.53	0.835
		RNA5916		**NA5916**	80	40	0.75	1.15
			RNA6916	**NA6916**	80	54	0.99	1.55
RLM9511526-1				**LM9511526-1**	85	26	0.48	0.75
	RNA49/82			—	—	—	0.57	—
RLM10012026-1				**LM10012026-1**	90	26	0.505	0.81
	RNA4917			**NA4917**	85	35	0.68	1.25
		RNA5917		**NA5917**	85	46	0.99	1.75
			RNA6917	**NA6917**	85	63	1.2	2.25
	RNA4918			**NA4918**	90	35	0.72	1.35
		RNA5918		**NA5918**	90	46	1.05	1.85
			RNA6918	**NA6918**	90	63	1.35	2.45
RLM11013030-1				—	—	—	0.635	—
	RNA4919			**NA4919**	95	35	0.74	1.4
RLM11013040-1				—	—	—	0.85	—
		RNA5919		**NA5919**	95	46	1.15	2.0
			RNA6919	**NA6919**	95	63	1.5	2.65
	RNA4920			**NA4920**	100	40	1.15	1.95
		RNA5920		**NA5920**	100	54	1.8	2.85

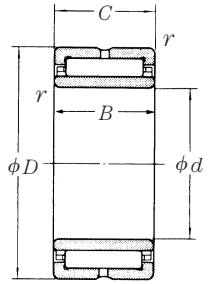
Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

RNA • NA

Inscribed Circle Diameter 120 — 265 mm



RNA



NA

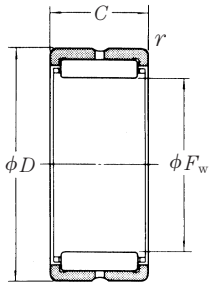
Boundary Dimensions (mm)				Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min⁻¹)
F_w	D	B, C	r min	C_r	C_{or}	C_r	C_{or}	Oil
120	140	30	1	99 500	214 000	10 100	21 900	3 400
125	150	40	1.1	149 000	315 000	15 200	32 500	3 200
	150	54	1.1	200 000	460 000	20 300	47 000	3 200
130	150	30	1	105 000	238 000	10 700	24 300	3 200
135	165	45	1.1	192 000	395 000	19 600	40 500	3 000
	165	60	1.1	253 000	565 000	25 800	58 000	3 000
145	165	35	1.1	127 000	315 000	12 900	32 000	2 800
150	180	50	1.5	228 000	515 000	23 200	52 500	2 800
	180	67	1.5	299 000	725 000	30 500	74 000	2 800
155	175	35	1.1	133 000	340 000	13 600	35 000	2 600
160	190	50	1.5	235 000	545 000	24 000	55 500	2 600
	190	67	1.5	310 000	775 000	31 500	79 000	2 600
165	190	40	1.1	180 000	440 000	18 300	45 000	2 400
170	210	60	2	315 000	645 000	32 000	66 000	2 400
175	200	40	1.1	184 000	465 000	18 700	47 000	2 200
180	220	60	2	325 000	695 000	33 500	71 000	2 200
185	215	45	1.1	224 000	540 000	22 900	55 000	2 200
190	230	60	2	340 000	745 000	34 500	76 000	2 200
195	225	45	1.1	230 000	565 000	23 500	58 000	2 000
205	250	69	2	400 000	940 000	41 000	96 000	2 000
210	240	50	1.5	268 000	705 000	27 300	72 000	1 900
215	260	69	2	415 000	1 000 000	42 000	102 000	1 900
220	250	50	1.5	274 000	740 000	27 900	75 500	1 800
225	280	80	2.1	525 000	1 140 000	53 500	116 000	1 900
240	270	50	1.5	286 000	805 000	29 100	82 000	1 700
245	300	80	2.1	545 000	1 230 000	55 500	125 000	1 700
265	300	60	2	375 000	1 070 000	38 500	109 000	1 500
	320	80	2.1	590 000	1 400 000	60 500	142 000	1 600

Bearing Numbers				Boundary Dimensions (mm)	Mass (kg) approx	
Without Inner Ring			With Inner Ring		Without Inner Ring	With Inner Ring
RNA48	RNA49	RNA59	NA	d		
RNA4822			**NA4822**	110	0.67	1.1
	RNA4922		**NA4922**	110	1.25	2.1
		RNA5922	**NA5922**	110	1.95	3.05
RNA4824			**NA4824**	120	0.71	1.15
	RNA4924		**NA4924**	120	1.9	2.9
		RNA5924	**NA5924**	120	2.7	4.05
RNA4826			**NA4826**	130	0.92	1.8
	RNA4926		**NA4926**	130	2.3	4.0
		RNA5926	**NA5926**	130	3.3	5.55
RNA4828			**NA4828**	140	0.98	1.9
	RNA4928		**NA4928**	140	2.45	4.25
		RNA5928	**NA5928**	140	3.55	6.0
RNA4830			**NA4830**	150	1.6	2.75
	RNA4930		**NA4930**	150	3.9	6.25
RNA4832			**NA4832**	160	1.75	2.95
	RNA4932		**NA4932**	160	4.1	6.6
RNA4834			**NA4834**	170	2.55	4.0
	RNA4934		**NA4934**	170	4.3	6.95
RNA4836			**NA4836**	180	2.65	4.2
	RNA4936		**NA4936**	180	6.2	10
RNA4838			**NA4838**	190	3.2	5.6
	RNA4938		**NA4938**	190	6.45	10.5
RNA4840			**NA4840**	200	3.35	5.9
	RNA4940		**NA4940**	200	9.7	15
RNA4844			**NA4844**	220	3.65	6.45
	RNA4944		**NA4944**	220	10	15.5
RNA4848			**NA4848**	240	5.45	10
	RNA4948		**NA4948**	240	11.5	17.5

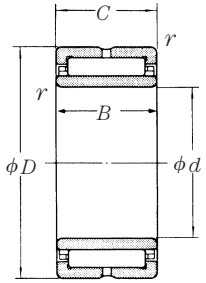
Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

RNA • NA

Inscribed Circle Diameter 285 — 490 mm



RNA



NA

Boundary Dimensions (mm)				Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})
F_w	D	B, C	r min	C_r	C_{or}	C_r	C_{or}	Oil
285	320	60	2	395 000	1 160 000	40 000	118 000	1 400
290	360	100	2.1	870 000	1 910 000	88 500	194 000	1 400
305	350	69	2	510 000	1 390 000	52 000	142 000	1 300
310	380	100	2.1	905 000	2 050 000	92 500	209 000	1 300
330	380	80	2.1	660 000	1 810 000	67 500	185 000	1 200
340	420	118	3	1 150 000	2 630 000	117 000	269 000	1 200
350	400	80	2.1	675 000	1 900 000	69 000	194 000	1 100
360	440	118	3	1 190 000	2 820 000	122 000	288 000	1 100
370	420	80	2.1	690 000	1 990 000	70 500	203 000	1 100
380	460	118	3	1 240 000	3 000 000	126 000	305 000	1 100
390	440	80	2.1	705 000	2 080 000	72 000	212 000	1 000
400	480	118	3	1 280 000	3 200 000	130 000	325 000	1 000
415	480	100	2.1	1 030 000	2 940 000	105 000	299 000	1 000
430	520	140	4	1 550 000	3 750 000	158 000	385 000	950
450	540	140	4	1 600 000	4 000 000	164 000	410 000	900
470	560	140	4	1 660 000	4 250 000	170 000	435 000	900
490	600	160	4	1 980 000	4 750 000	202 000	485 000	850

Bearing Numbers				Boundary Dimensions (mm)	Mass (kg) approx	
Without Inner Ring			With Inner Ring			
RNA48	RNA49	RNA59	NA	d	Without Inner Ring	With Inner Ring
RNA4852			**NA4852**	260	5.9	11
	RNA4952		**NA4952**	260	19.5	29.5
RNA4856			**NA4856**	280	9.5	15.5
	RNA4956		**NA4956**	280	20.5	31
RNA4860			**NA4860**	300	13	22
	RNA4960		**NA4960**	300	30	48.5
RNA4864			**NA4864**	320	13.5	23.5
	RNA4964		**NA4964**	320	32	51.5
RNA4868			**NA4868**	340	14	24.5
	RNA4968		**NA4968**	340	33.5	54
RNA4872			**NA4872**	360	15	26
	RNA4972		**NA4972**	360	35.5	57
RNA4876			**NA4876**	380	25.5	42.5
	RNA4976		**NA4976**	380	50.5	85.5
	RNA4980		**NA4980**	400	52.5	89
	RNA4984		**NA4984**	420	54.5	92.5
	RNA4988		**NA4988**	440	81.5	125

Remarks 1. For information on inner rings, refer to page B129.
2. If a full complement roller bearing is required, please contact NSK.

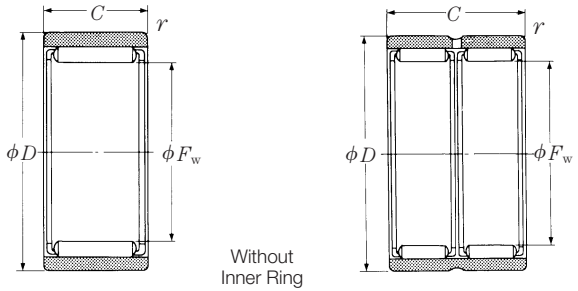
RNAF (Single-Row, Without Inner Ring)

RNAFW (Double-Row, Without Inner Ring)

NAF (Single-Row, With Inner Ring)

NAFW (Double-Row, With Inner Ring)

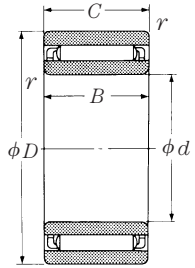
Inscribed Circle Diameter 10 — 30 mm



RNAF

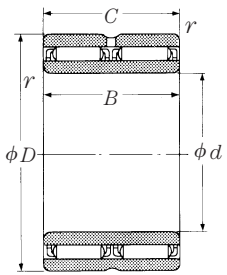
Without Inner Ring

RNAFW



NAF

With Inner Ring



NAFW

Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
10	17	10	0.3	3 950	4 300	405	435	45 000
	20	12	0.3	7 700	6 650	785	680	48 000
12	22	12	0.3	9 150	8 650	935	885	38 000
14	22	13	0.3	8 050	9 750	820	995	32 000
	22	20	0.3	11 400	15 200	1 160	1 550	32 000
	26	12	0.3	9 850	9 800	1 010	1 000	32 000
15	23	13	0.3	8 400	10 500	860	1 070	28 000
	23	20	0.3	12 100	16 800	1 230	1 710	28 000
16	24	13	0.3	8 800	11 300	895	1 150	26 000
	24	20	0.3	12 600	18 000	1 290	1 840	26 000
	28	12	0.3	10 500	10 900	1 070	1 120	28 000
17	25	13	0.3	10 200	14 000	1 040	1 420	26 000
	25	20	0.3	13 100	19 300	1 340	1 970	26 000
18	26	13	0.3	9 450	12 900	965	1 310	24 000
	26	20	0.3	13 600	20 500	1 380	2 090	24 000
	30	12	0.3	11 700	13 000	1 190	1 320	24 000
	30	24	0.3	20 100	26 000	2 050	2 650	24 000
20	28	13	0.3	9 700	13 700	990	1 400	20 000
	28	26	0.3	16 600	27 500	1 690	2 800	20 000
	32	12	0.3	12 200	14 100	1 250	1 440	22 000
	32	24	0.3	20 900	28 200	2 140	2 880	22 000
22	30	13	0.3	10 300	15 300	1 050	1 560	19 000
	30	26	0.3	17 600	30 500	1 800	3 100	19 000
	35	16	0.3	16 600	19 500	1 700	1 980	20 000
	35	32	0.3	29 400	40 500	2 990	4 100	20 000
25	35	17	0.3	14 800	25 500	1 510	2 600	17 000
	35	26	0.3	19 400	36 000	1 970	3 650	17 000
	37	16	0.3	18 700	23 500	1 910	2 390	17 000
	37	32	0.3	32 000	47 000	3 250	4 800	17 000
28	40	16	0.3	20 200	26 800	2 060	2 730	15 000
	40	32	0.3	34 500	53 500	3 550	5 450	15 000
30	40	17	0.3	18 700	31 500	1 910	3 200	14 000
	40	26	0.3	24 000	43 000	2 450	4 400	14 000
	42	16	0.3	22 200	31 000	2 270	3 150	14 000
	42	32	0.3	38 000	62 000	3 900	6 300	14 000

Bearing Numbers	Boundary Dimensions (mm)		Bearing Numbers	Mass (kg) approx	
Without Inner Ring	d	B	With Inner Ring	Without Inner Ring	With Inner Ring
RNAF 101710	—	—	—	0.010	—
RNAF 102012	—	—	—	0.018	—
RNAF 122212	—	—	—	0.021	—
RNAF 142213	10	13	**NAF 102213**	0.019	0.027
RNAFW 142220	10	20	**NAFW 102220**	0.030	0.042
RNAF 142612	10	12	**NAF 102612**	0.027	0.034
RNAF 152313	—	—	—	0.021	—
RNAFW 152320	—	—	—	0.032	—
RNAF 162413	12	13	**NAF 122413**	0.022	0.031
RNAFW 162420	12	20	**NAFW 122420**	0.034	0.048
RNAF 162812	12	12	**NAF 122812**	0.032	0.040
RNAF 172513	—	—	—	0.022	—
RNAFW 172520	—	—	—	0.035	—
RNAF 182613	—	—	—	0.024	—
RNAFW 182620	—	—	—	0.037	—
RNAF 183012	—	—	—	0.036	—
RNAFW 183024	—	—	—	0.072	—
RNAF 202813	15	13	**NAF 152813**	0.026	0.040
RNAFW 202826	15	26	**NAFW 152826**	0.051	0.079
RNAF 203212	15	12	**NAF 153212**	0.038	0.050
RNAFW 203224	—	—	—	0.076	—
RNAF 223013	17	13	**NAF 173013**	0.028	0.044
RNAFW 223026	17	26	**NAFW 173026**	0.056	0.087
RNAF 223516	17	16	**NAF 173516**	0.061	0.080
RNAFW 223532	17	32	**NAFW 173532**	0.12	0.16
RNAF 253517	20	17	**NAF 203517**	0.054	0.077
RNAFW 253526	20	26	**NAFW 203526**	0.080	0.115
RNAF 253716	20	16	**NAF 203716**	0.060	0.082
RNAFW 253732	20	32	**NAFW 203732**	0.12	0.16
RNAF 284016	—	—	—	0.067	—
RNAFW 284032	—	—	—	0.13	—
RNAF 304017	25	17	**NAF 254017**	0.064	0.092
RNAFW 304026	25	26	**NAFW 254026**	0.096	0.14
RNAF 304216	25	16	**NAF 254216**	0.069	0.095
RNAFW 304232	25	32	**NAFW 254232**	0.14	0.19

Remarks For information on inner rings, refer to page B129.

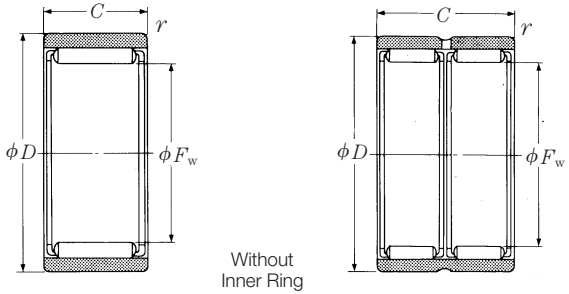
RNAF (Single-Row, Without Inner Ring)

RNAFW (Double-Row, Without Inner Ring)

NAF (Single-Row, With Inner Ring)

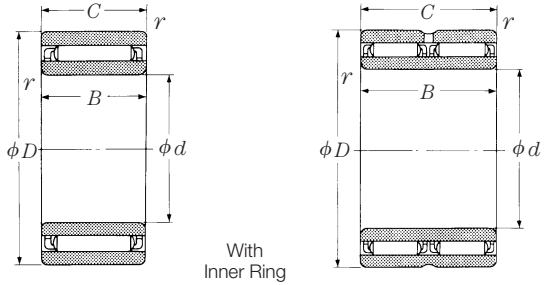
NAFW (Double-Row, With Inner Ring)

Inscribed Circle Diameter 35 — 100 mm



Without Inner Ring

RNAF RNAFW



With Inner Ring

NAF NAFW

Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Oil
35	45	13	0.3	14 900	24 600	1 520	2 500	12 000
	45	17	0.3	20 500	37 000	2 090	3 750	12 000
	45	26	0.3	25 600	49 000	2 610	5 000	12 000
	47	16	0.3	22 700	33 500	2 310	3 400	12 000
	47	32	0.3	39 000	66 500	3 950	6 800	12 000
40	50	17	0.3	21 000	40 000	2 150	4 050	10 000
	50	34	0.3	36 000	80 000	3 700	8 150	10 000
	55	20	0.3	32 500	49 500	3 300	5 050	10 000
	55	40	0.3	55 500	99 500	5 650	10 100	10 000
45	55	17	0.3	21 600	43 000	2 200	4 350	9 000
	55	34	0.3	37 000	85 500	3 800	8 750	9 000
	62	20	0.3	34 000	55 000	3 450	5 600	9 500
	62	40	0.3	58 000	110 000	5 950	11 200	9 500
50	62	20	0.3	26 900	59 000	2 750	6 050	8 000
	62	40	0.3	46 000	118 000	4 700	12 100	8 000
	65	20	0.3	38 500	67 500	3 950	6 900	8 500
	65	40	0.6	66 000	135 000	6 750	13 800	8 500
55	68	20	0.6	27 500	62 500	2 800	6 350	7 500
	68	40	0.6	47 000	125 000	4 800	12 700	7 500
	72	20	0.6	40 000	73 000	4 100	7 450	7 500
	72	40	0.6	68 500	146 000	7 000	14 900	7 500
60	78	20	1	41 500	78 000	4 200	7 950	6 700
	78	40	1	71 000	156 000	7 250	15 900	6 700
65	85	30	1	61 000	132 000	6 200	13 400	6 300
	85	60	1	104 000	263 000	10 600	26 800	6 300
70	90	30	1	63 000	140 000	6 400	14 300	6 000
	90	60	1	108 000	281 000	11 000	28 600	6 000
75	95	30	1	65 000	151 000	6 650	15 400	5 600
	95	60	1	112 000	300 000	11 400	31 000	5 600
80	100	30	1	69 000	166 000	7 050	17 000	5 000
	100	60	1	119 000	335 000	12 100	34 000	5 000
85	105	30	1	71 000	176 000	7 250	17 900	4 800
90	110	30	1	70 000	177 000	7 150	18 000	4 500
95	115	30	1	69 500	177 000	7 100	18 100	4 300
100	120	30	1	75 500	201 000	7 700	20 500	4 000

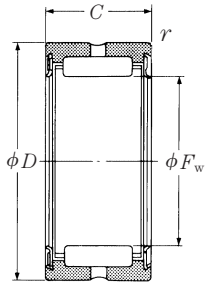
Bearing Numbers	Boundary Dimensions (mm)		Bearing Numbers	Mass (kg) approx	
Without Inner Ring	d	B	With Inner Ring	Without Inner Ring	With Inner Ring
RNAF 354513	30	13	**NAF 304513**	0.053	0.078
RNAF 354517	30	17	**NAF 304517**	0.073	0.105
RNAFW 354526	30	26	**NAFW 304526**	0.105	0.155
RNAF 354716	30	16	**NAF 304716**	0.078	0.11
RNAFW 354732	30	32	**NAFW 304732**	0.16	0.22
RNAF 405017	35	17	**NAF 355017**	0.084	0.12
RNAFW 405034	35	34	**NAFW 355034**	0.165	0.24
RNAF 405520	35	20	**NAF 355520**	0.15	0.195
RNAFW 405540	35	40	**NAFW 355540**	0.295	0.385
RNAF 455517	40	17	**NAF 405517**	0.092	0.135
RNAFW 455534	40	34	**NAFW 405534**	0.18	0.265
RNAF 456220	40	20	**NAF 406220**	0.165	0.215
RNAFW 456240	40	40	**NAFW 406240**	0.325	0.43
RNAF 506220	45	20	**NAF 456220**	0.14	0.195
RNAFW 506240	45	40	**NAFW 456240**	0.28	0.395
RNAF 506520	—	—	—	0.175	—
RNAFW 506540	—	—	—	0.36	—
RNAF 556820	50	20	**NAF 506820**	0.165	0.23
RNAFW 556840	50	40	**NAFW 506840**	0.34	0.465
RNAF 557220	45	20	**NAF 457220**	0.225	0.345
RNAFW 557240	45	40	**NAFW 457240**	0.455	0.695
RNAF 607820	50	20	**NAF 507820**	0.265	0.40
RNAFW 607840	50	40	**NAFW 507840**	0.535	0.80
RNAF 658530	55	30	**NAF 558530**	0.495	0.71
RNAFW 658560	55	60	**NAFW 558560**	0.99	1.4
RNAF 709030	60	30	**NAF 609030**	0.53	0.765
RNAFW 709060	60	60	**NAFW 609060**	1.05	1.5
RNAF 759530	65	30	**NAF 659530**	0.575	0.825
RNAFW 759560	65	60	**NAFW 659560**	1.15	1.65
RNAF 8010030	70	30	**NAF 7010030**	0.60	0.87
RNAFW 8010060	70	60	**NAFW 7010060**	1.2	1.7
RNAF 8510530	75	30	**NAF 7510530**	0.625	0.915
RNAF 9011030	80	30	**NAF 8011030**	0.665	0.97
RNAF 9511530	85	30	**NAF 8511530**	0.695	1.0
RNAF 10012030	90	30	**NAF 9012030**	0.73	1.05

Remarks For information on inner rings, refer to page B129.

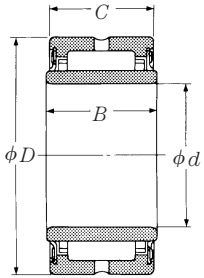
RNA···TT (Sealed, Without Inner Ring)

NA···TT (Sealed, With Inner Ring)

Inscribed Circle Diameter 14 — 52 mm



RNA···TT



NA···TT

Boundary Dimensions (mm)				Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})	Bearing Numbers
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Grease	Without Inner Ring
14	22	13	0.3	7 350	7 500	750	765	13 000	**RNA4900TT**
16	24	13	0.3	8 100	8 850	825	900	11 000	**RNA4901TT**
	24	22	0.3	15 400	20 300	1 580	2 070	11 000	**RNA6901TT**
18	26	13	0.3	8 400	9 600	860	975	10 000	**RNA49/14TT**
20	28	13	0.3	8 700	10 300	890	1 050	9 500	**RNA4902TT**
	28	23	0.3	17 500	25 300	1 780	2 580	9 500	**RNA6902TT**
22	30	13	0.3	9 350	11 600	955	1 190	8 500	**RNA4903TT**
	30	23	0.3	18 800	28 500	1 920	2 910	8 500	**RNA6903TT**
25	37	17	0.3	16 800	18 700	1 710	1 910	7 500	**RNA4904TT**
	37	30	0.3	33 000	44 000	3 350	4 500	7 500	**RNA6904TT**
28	39	17	0.3	17 800	22 900	1 820	2 330	6 700	**RNA49/22TT**
	39	30	0.3	33 500	51 500	3 400	5 250	6 700	**RNA69/22TT**
30	42	17	0.3	19 900	24 600	2 030	2 510	6 300	**RNA4905TT**
	42	30	0.3	30 000	41 500	3 100	4 250	6 300	**RNA6905TT**
32	45	17	0.3	18 900	23 400	1 930	2 390	5 600	**RNA49/28TT**
	45	30	0.3	37 000	55 500	3 750	5 650	5 600	**RNA69/28TT**
35	47	17	0.3	20 400	26 400	2 080	2 690	5 300	**RNA4906TT**
	47	30	0.3	40 000	62 500	4 050	6 350	5 300	**RNA6906TT**
40	52	20	0.6	28 000	41 500	2 860	4 250	4 500	**RNA49/32TT**
	52	36	0.6	51 500	91 000	5 250	9 250	4 500	**RNA69/32TT**
42	55	20	0.6	29 700	45 500	3 050	4 650	4 500	**RNA4907TT**
	55	36	0.6	53 000	95 500	5 400	9 700	4 500	**RNA6907TT**
45	58	20	0.6	30 500	47 500	3 100	4 850	4 000	**RNA49/38TT**
48	62	22	0.6	34 500	53 500	3 550	5 450	3 800	**RNA4908TT**
	62	40	0.6	67 000	124 000	6 800	12 700	3 800	**RNA6908TT**
50	65	22	0.6	36 500	58 000	3 750	5 950	3 800	**RNA49/42TT**
52	68	22	0.6	36 500	58 500	3 700	5 950	3 600	**RNA4909TT**
	68	40	0.6	70 500	136 000	7 150	13 800	3 600	**RNA6909TT**

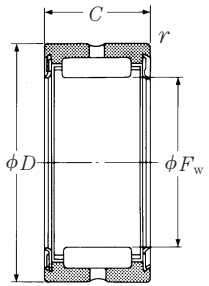
Numbers	Boundary Dimensions (mm)		Mass (kg) approx	
With Inner Ring	d	B	Without Inner Ring	With Inner Ring
NA4900TT	10	14	0.016	0.025
NA4901TT	12	14	0.018	0.028
NA6901TT	12	23	0.030	0.046
—	—	—	0.020	—
NA4902TT	15	14	0.021	0.036
NA6902TT	15	24	0.039	0.065
NA4903TT	17	14	0.023	0.039
NA6903TT	17	24	0.041	0.069
NA4904TT	20	18	0.055	0.078
NA6904TT	20	31	0.098	0.14
NA49/22TT	22	18	0.056	0.088
NA69/22TT	22	31	0.096	0.15
NA4905TT	25	18	0.063	0.093
NA6905TT	25	31	0.11	0.165
NA49/28TT	28	18	0.076	0.10
NA69/28TT	28	31	0.13	0.18
NA4906TT	30	18	0.072	0.105
NA6906TT	30	31	0.13	0.19
NA49/32TT	32	21	0.092	0.165
NA69/32TT	32	37	0.17	0.295
NA4907TT	35	21	0.11	0.175
NA6907TT	35	37	0.20	0.32
—	—	—	0.12	—
NA4908TT	40	23	0.15	0.245
NA6908TT	40	41	0.265	0.44
—	—	—	0.17	—
NA4909TT	45	23	0.19	0.285
NA6909TT	45	41	0.335	0.50

Remarks 1. For information on inner rings, refer to page B129.
2. Standard greaes is packed in sealed solid bearings unless otherwise specified.

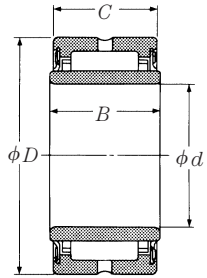
RNA···TT (**Sealed, Without Inner Ring**)

NA···TT (**Sealed, With Inner Ring**)

Inscribed Circle Diameter 55 — 125 mm



RNA···TT



NA···TT

Boundary Dimensions (mm)				Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min⁻¹)	Bearing Numbers
F_w	D	C	r min	C_r	C_{or}	C_r	C_{or}	Grease	Without Inner Ring
55	70	22	0.6	38 500	63 500	3 900	6 450	3 400	**RNA49/48TT**
58	72	22	0.6	38 000	63 500	3 850	6 500	3 200	**RNA4910TT**
	72	40	0.6	73 000	148 000	7 450	15 100	3 200	**RNA6910TT**
63	80	25	1	47 500	76 000	4 850	7 750	3 000	**RNA4911TT**
	80	45	1	90 000	172 000	9 200	17 500	3 000	**RNA6911TT**
68	85	25	1	50 000	82 500	5 100	8 450	2 800	**RNA4912TT**
	85	45	1	94 500	187 000	9 600	19 100	2 800	**RNA6912TT**
72	90	25	1	52 500	89 500	5 350	9 100	2 600	**RNA4913TT**
	90	45	1	99 000	203 000	10 100	20 700	2 600	**RNA6913TT**
80	100	30	1	70 000	120 000	7 150	12 200	2 200	**RNA4914TT**
	100	54	1	130 000	264 000	13 200	26 900	2 200	**RNA6914TT**
85	105	30	1	77 000	139 000	7 850	14 100	2 200	**RNA4915TT**
	105	54	1	136 000	286 000	13 800	29 100	2 200	**RNA6915TT**
90	110	30	1	76 500	139 000	7 800	14 200	2 000	**RNA4916TT**
	110	54	1	141 000	305 000	14 400	31 500	2 000	**RNA6916TT**
100	120	35	1.1	94 500	189 000	9 600	19 300	1 900	**RNA4917TT**
	120	63	1.1	167 000	395 000	17 100	40 000	1 900	**RNA6917TT**
105	125	35	1.1	98 000	201 000	10 000	20 500	1 800	**RNA4918TT**
	125	63	1.1	173 000	420 000	17 700	43 000	1 800	**RNA6918TT**
110	130	35	1.1	101 000	214 000	10 300	21 800	1 700	**RNA4919TT**
	130	63	1.1	181 000	450 000	18 500	46 000	1 700	**RNA6919TT**
115	140	40	1.1	131 000	261 000	13 400	26 600	1 600	**RNA4920TT**
125	150	40	1.1	136 000	279 000	13 800	28 500	1 500	**RNA4922TT**

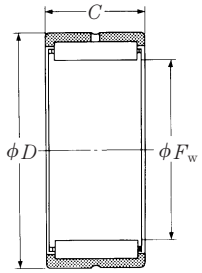
Bearing Numbers	Boundary Dimensions (mm)		Mass (kg) approx	
With Inner Ring	d	B	Without Inner Ring	With Inner Ring
—	—	—	0.18	—
NA4910TT	50	23	0.18	0.30
NA6910TT	50	41	0.32	0.535
NA4911TT	55	26	0.26	0.405
NA6911TT	55	46	0.475	0.735
NA4912TT	60	26	0.28	0.44
NA6912TT	60	46	0.485	0.765
NA4913TT	65	26	0.32	0.47
NA6913TT	65	46	0.53	0.795
NA4914TT	70	31	0.47	0.75
NA6914TT	70	55	0.89	1.4
NA4915TT	75	31	0.50	0.80
NA6915TT	75	55	0.96	1.5
NA4916TT	80	31	0.53	0.845
NA6916TT	80	55	0.99	1.55
NA4917TT	85	36	0.68	1.25
NA6917TT	85	64	1.2	2.25
NA4918TT	90	36	0.72	1.35
NA6918TT	90	64	1.35	2.45
NA4919TT	95	36	0.74	1.4
NA6919TT	95	64	1.5	2.65
NA4920TT	100	41	1.15	1.95
NA4922TT	110	41	1.25	2.1

Remarks 1. For information on inner rings, refer to page B129.
2. Standard greaes is packed in sealed solid bearings unless otherwise specified.

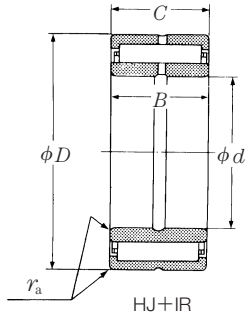
HJ (Single-Row, Without Inner Ring)

HJ + IR (Single-Row, With Inner Ring)

Inscribed Circle Diameter 15.875 — 44.450 mm



HJ



HJ+IR

* Fillet Radius of Shaft, Housing

Bearing Numbers Without Inner Ring	Boundary Dimensions (mm, inch) F_w		D		C		Basic Load Ratings (N) C_r	C_{or}	{kgf} C_r	C_{or}	Limiting Speeds (min^{-1}) Oil
15.875	0.6250	28.575	1.1250	19.05	0.7500	18 600	19 400	1 890	1 980	30 000
19.050	0.7500	31.750	1.2500	19.05	0.7500	19 800	21 900	2 020	2 230	24 000
		31.750	1.2500	25.40	1.0000	26 800	32 000	2 730	3 300	24 000
22.225	0.8750	34.925	1.3750	19.05	0.7500	22 100	26 200	2 260	2 670	20 000
		34.925	1.3750	25.40	1.0000	29 900	38 500	3 050	3 950	20 000
25.400	1.0000	38.100	1.5000	19.05	0.7500	24 200	30 500	2 470	3 100	17 000
		38.100	1.5000	25.40	1.0000	32 500	45 000	3 350	4 600	17 000
28.575	1.1250	41.275	1.6250	25.40	1.0000	35 500	51 500	3 600	5 250	15 000
		41.275	1.6250	31.75	1.2500	44 000	68 000	4 500	6 950	15 000
31.750	1.2500	44.450	1.7500	25.40	1.0000	36 500	55 000	3 700	5 600	13 000
		44.450	1.7500	31.75	1.2500	45 500	72 500	4 600	7 400	13 000
34.925	1.3750	47.625	1.8750	25.40	1.0000	38 500	61 000	3 950	6 250	12 000
		47.625	1.8750	31.75	1.2500	48 000	81 000	4 900	8 250	12 000
38.100	1.5000	52.388	2.0625	25.40	1.0000	46 000	68 500	4 650	7 000	11 000
		52.388	2.0625	31.75	1.2500	57 000	91 000	5 850	9 300	11 000
41.275	1.6250	55.562	2.1875	25.40	1.0000	47 000	72 500	4 800	7 400	10 000
		55.562	2.1875	31.75	1.2500	58 500	96 500	6 000	9 850	10 000
44.450	1.7500	58.738	2.3125	25.40	1.0000	48 000	76 500	4 900	7 800	9 500
		58.738	2.3125	31.75	1.2500	60 000	102 000	6 150	10 400	9 500

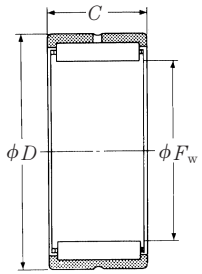
Bearing Numbers Without Inner Ring	Boundary Dimensions (mm, inch) d		B		Matching Inner Rings	* (mm) r_a max	Mass (kg) approx Without Inner Ring	With Inner Ring
HJ-101812	9.525	0.3750	19.05	0.7500	**IR-061012**	0.6	0.050	0.068
HJ-122012	12.700	0.5000	19.05	0.7500	**IR-081212**	1	0.054	0.081
HJ-122016	12.700	0.5000	25.40	1.0000	**IR-081216**	1	0.073	0.11
HJ-142212	15.875	0.6250	19.05	0.7500	**IR-101412**	1	0.064	0.090
	17.462	0.6875	19.05	0.7500	**IR-111412**	1	0.064	0.087
HJ-142216	15.875	0.6250	25.40	1.0000	**IR-101416**	1	0.082	0.115
HJ-162412	19.050	0.7500	19.05	0.7500	**IR-121612**	1	0.068	0.10
HJ-162416	19.050	0.7500	25.40	1.0000	**IR-121616**	1	0.091	0.135
	20.638	0.8125	25.40	1.0000	**IR-131616**	1	0.091	0.125
HJ-182616	22.225	0.8750	25.40	1.0000	**IR-141816**	1	0.10	0.15
	23.812	0.9375	25.40	1.0000	**IR-151816**	1	0.10	0.14
HJ-182620	22.225	0.8750	31.75	1.2500	**IR-141820**	1	0.13	0.195
	23.812	0.9375	31.75	1.2500	**IR-151820**	1	0.13	0.18
HJ-202816	25.400	1.0000	25.40	1.0000	**IR-162016**	1	0.11	0.17
HJ-202820	25.400	1.0000	31.75	1.2500	**IR-162020**	1	0.14	0.215
HJ-223016	28.575	1.1250	25.40	1.0000	**IR-182216**	1	0.12	0.185
HJ-223020	28.575	1.1250	31.75	1.2500	**IR-182220**	1	0.155	0.23
HJ-243316	31.750	1.2500	25.40	1.0000	**IR-202416**	1.5	0.155	0.23
HJ-243320	30.162	1.1875	31.75	1.2500	**IR-192420**	1.5	0.195	0.30
	31.750	1.2500	31.75	1.2500	**IR-202420**	1.5	0.195	0.285
HJ-263516	33.338	1.3125	25.40	1.0000	**IR-212616**	1.5	0.16	0.255
HJ-263520	33.338	1.3125	31.75	1.2500	**IR-212620**	1.5	0.20	0.32
	34.925	1.3750	31.75	1.2500	**IR-222620**	1.5	0.20	0.30
HJ-283716	36.512	1.4375	25.40	1.0000	**IR-232816**	1.5	0.17	0.27
	38.100	1.5000	25.40	1.0000	**IR-242816**	1.5	0.17	0.265
HJ-283720	34.925	1.3750	31.75	1.2500	**IR-222820**	1.5	0.215	0.36
	36.512	1.4375	31.75	1.2500	**IR-232820**	1.5	0.215	0.34
	38.100	1.5000	31.75	1.2500	**IR-242820**	1.5	0.215	0.315

Remarks
1. For bearings with inner rings, the inner ring number is written separately from the Bearing number.
   Example: HJ-202816 + IR-162016
2. If a full complement roller bearing is required, please contact NSK.

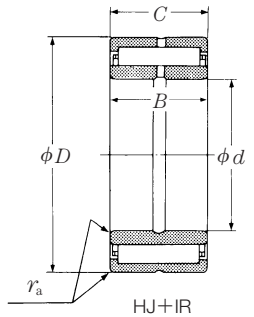
HJ (Single-Row, Without Inner Ring)

HJ + IR (Single-Row, With Inner Ring)

Inscribed Circle Diameter 47.625 — 101.600 mm



HJ



HJ+IR

* Fillet Radius of Shaft, Housing

Bearing Numbers Without Inner Ring	Boundary Dimensions (mm, inch) F_w	D		C		Basic Load Ratings (N) C_r	C_{or}	{kgf} C_r	C_{or}	Limiting Speeds (min^{-1}) Oil
47.625	1.8750	61.912	2.4375	31.75	1.2500	63 500	112 000	6 500	11 400	9 000
50.800	2.0000	65.088	2.5625	25.40	1.0000	52 000	88 000	5 300	9 000	8 500
		65.088	2.5625	31.75	1.2500	65 000	117 000	6 600	12 000	8 500
57.150	2.2500	76.200	3.0000	38.10	1.5000	89 000	161 000	9 100	16 400	7 500
		76.200	3.0000	44.45	1.7500	103 000	194 000	10 500	19 800	7 500
63.500	2.5000	82.550	3.2500	38.10	1.5000	96 000	183 000	9 800	18 700	6 700
		82.550	3.2500	44.45	1.7500	111 000	221 000	11 300	22 600	6 700
69.850	2.7500	88.900	3.5000	25.40	1.0000	66 000	116 000	6 700	11 800	6 000
		88.900	3.5000	38.10	1.5000	100 000	199 000	10 200	20 300	6 000
		88.900	3.5000	44.45	1.7500	116 000	240 000	11 800	24 500	6 000
76.200	3.0000	95.250	3.7500	38.10	1.5000	106 000	221 000	10 800	22 600	5 600
		95.250	3.7500	44.45	1.7500	123 000	268 000	12 600	27 300	5 600
82.550	3.2500	107.950	4.2500	44.45	1.7500	161 000	300 000	16 400	30 500	5 000
		107.950	4.2500	50.80	2.0000	182 000	350 000	18 600	36 000	5 000
88.900	3.5000	114.300	4.5000	50.80	2.0000	186 000	370 000	18 900	37 500	4 800
95.250	3.7500	120.650	4.7500	50.80	2.0000	194 000	400 000	19 800	40 500	4 500
101.600	4.0000	127.000	5.0000	50.80	2.0000	202 000	430 000	20 600	43 500	4 000

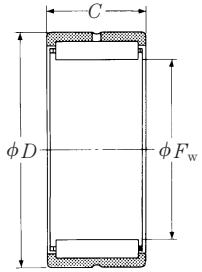
Bearing Numbers Without Inner Ring	Boundary Dimensions (mm, inch) d		B		Matching Inner Rings	* (mm) r_a max	Mass (kg) approx Without Inner Ring	With Inner Ring
HJ-303920	39.688	1.5625	31.75	1.2500	**IR-253020**	1.5	0.225	0.36
HJ-324116	42.862	1.6875	25.40	1.0000	**IR-273216**	1.5	0.185	0.305
HJ-324120	38.100	1.5000	31.75	1.2500	**IR-243220**	1.5	0.235	0.455
	39.688	1.5625	31.75	1.2500	**IR-253220**	1.5	0.235	0.43
	41.275	1.6250	31.75	1.2500	**IR-263220**	1.5	0.235	0.405
	42.862	1.6875	31.75	1.2500	**IR-273220**	1.5	0.235	0.38
HJ-364824	44.450	1.7500	38.10	1.5000	**IR-283624**	1.5	0.45	0.755
HJ-364828	44.450	1.7500	44.45	1.7500	**IR-283628**	1.5	0.525	0.88
HJ-405224	49.212	1.9375	38.10	1.5000	**IR-314024**	2	0.525	0.90
	50.800	2.0000	38.10	1.5000	**IR-324024**	2	0.525	0.865
HJ-405228	49.212	1.9375	44.45	1.7500	**IR-314028**	2	0.575	1.0
	50.800	2.0000	44.45	1.7500	**IR-324028**	2	0.575	0.975
HJ-445616	—	—	—	—	—	2	0.485	—
HJ-445624	57.150	2.2500	38.10	1.5000	**IR-364424**	2	0.595	0.97
HJ-445628	55.562	2.1875	44.45	1.7500	**IR-354428**	2	0.65	1.15
	57.150	2.2500	44.45	1.7500	**IR-364428**	2	0.65	1.1
HJ-486024	63.500	2.5000	38.10	1.5000	**IR-404824**	2	0.61	1.05
HJ-486028	60.325	2.3750	44.45	1.7500	**IR-384828**	2	0.68	1.25
	63.500	2.5000	44.45	1.7500	**IR-404828**	2	0.68	1.15
HJ-526828	69.850	2.7500	44.45	1.7500	**IR-445228**	2	1.05	1.55
HJ-526832	69.850	2.7500	50.80	2.0000	**IR-445232**	2	1.2	1.8
HJ-567232	74.612	2.9375	50.80	2.0000	**IR-475632**	2	1.2	1.95
	76.200	3.0000	50.80	2.0000	**IR-485632**	2	1.2	1.9
HJ-607632	79.375	3.1250	50.80	2.0000	**IR-506032**	2.5	1.3	2.2
	82.550	3.2500	50.80	2.0000	**IR-526032**	2.5	1.3	2.0
HJ-648032	82.550	3.2500	50.80	2.0000	**IR-526432**	2.5	1.4	2.5
	85.725	3.3750	50.80	2.0000	**IR-546432**	2.5	1.4	2.3
	88.900	3.5000	50.80	2.0000	**IR-566432**	2.5	1.4	2.15

Remarks
1. For bearings with inner rings, the inner ring number is written separately from the Bearing number.
   Example: HJ-202816 + IR-162016
2. If a full complement roller bearing is required, please contact NSK.

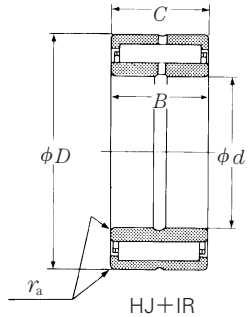
HJ (Single-Row, Without Inner Ring)

HJ + IR (Single-Row, With Inner Ring)

Inscribed Circle Diameter 107.950 — 234.950 mm



HJ



HJ+IR

* Fillet Radius of Shaft, Housing

Bearing Numbers Without Inner Ring	Boundary Dimensions (mm, inch) F_w		D		C		Basic Load Ratings (N) C_r	C_{or}	{kgf} C_r	C_{or}	Limiting Speeds (min^{-1}) Oil
107.950	4.2500	133.350	5.2500	50.80	2.0000	205 000	445 000	20 900	45 500	3 800
114.300	4.5000	152.400	6.0000	57.15	2.2500	290 000	525 000	29 600	53 500	3 800
		152.400	6.0000	63.50	2.5000	325 000	600 000	33 000	61 500	3 800
127.000	5.0000	165.100	6.5000	50.80	2.0000	279 000	515 000	28 400	52 500	3 400
		165.100	6.5000	57.15	2.2500	315 000	600 000	32 000	61 000	3 400
		165.100	6.5000	63.50	2.5000	350 000	685 000	35 500	70 000	3 400
139.700	5.5000	177.800	7.0000	63.50	2.5000	350 000	715 000	36 000	73 000	3 000
		177.800	7.0000	76.20	3.0000	415 000	890 000	42 500	91 000	3 000
146.050	5.7500	184.150	7.2500	76.20	3.0000	425 000	925 000	43 500	94 500	2 800
152.400	6.0000	190.500	7.5000	63.50	2.5000	375 000	795 000	38 000	81 500	2 800
		190.500	7.5000	76.20	3.0000	445 000	995 000	45 000	101 000	2 800
165.100	6.5000	203.200	8.0000	63.50	2.5000	385 000	855 000	39 500	87 000	2 400
		203.200	8.0000	76.20	3.0000	460 000	1 060 000	46 500	109 000	2 400
184.150	7.2500	231.775	9.1250	76.20	3.0000	535 000	1 140 000	54 500	117 000	2 200
196.850	7.7500	244.475	9.6250	76.20	3.0000	555 000	1 230 000	56 500	125 000	2 000
209.550	8.2500	257.175	10.1250	76.20	3.0000	575 000	1 310 000	58 500	133 000	2 000
222.250	8.7500	269.875	10.6250	76.20	3.0000	590 000	1 390 000	60 500	142 000	1 900
234.950	9.2500	282.575	11.1250	76.20	3.0000	610 000	1 470 000	62 000	150 000	1 800

Bearing Numbers Without Inner Ring	Boundary Dimensions (mm, inch) d		B		Matching Inner Rings	* (mm) r_a max	Mass (kg) approx Without Inner Ring	With Inner Ring
HJ-688432	88.900	3.5000	50.80	2.0000	**IR-566832**	2.5	1.5	2.65
	95.250	3.7500	50.80	2.0000	**IR-606832**	2.5	1.5	2.5
HJ-729636	95.250	3.7500	57.15	2.2500	**IR-607236**	2.5	2.75	4.15
HJ-729640	95.250	3.7500	63.50	2.5000	**IR-607240**	2.5	3.05	4.6
HJ-8010432	—	—	—	—	—	2.5	2.4	—
HJ-8010436	101.600	4.0000	57.15	2.2500	**IR-648036**	2.5	2.9	4.95
	107.950	4.2500	57.15	2.2500	**IR-688036**	2.5	2.9	4.45
HJ-8010440	101.600	4.0000	63.50	2.5000	**IR-648040**	2.5	3.3	5.55
HJ-8811240	114.300	4.5000	63.50	2.5000	**IR-728840**	2.5	3.6	6.1
HJ-8811248	114.300	4.5000	76.20	3.0000	**IR-728848**	2.5	4.25	7.2
HJ-9211648	120.650	4.7500	76.20	3.0000	**IR-769248**	3	4.55	7.7
HJ-9612040	127.000	5.0000	63.50	2.5000	**IR-809640**	3	3.9	6.65
HJ-9612048	127.000	5.0000	76.20	3.0000	**IR-809648**	3	4.75	8.1
HJ-10412840	139.700	5.5000	63.50	2.5000	**IR-8810440**	3	4.15	7.2
HJ-10412848	139.700	5.5000	76.20	3.0000	**IR-8810448**	3	4.75	8.4
HJ-11614648	152.400	6.0000	76.20	3.0000	**IR-9611648**	3	7.1	12
HJ-12415448	165.100	6.5000	76.20	3.0000	**IR-10412448**	3	7.5	13
HJ-13216248	177.800	7.0000	76.20	3.0000	**IR-11213248**	3	7.95	13.5
HJ-14017048	190.500	7.5000	76.20	3.0000	**IR-12014048**	4	8.35	14.5
HJ-14817848	203.200	8.0000	76.20	3.0000	**IR-12814048**	4	8.6	15

Remarks 1. For bearings with inner rings, the inner ring number is written separately from the Bearing number.
Example: HJ-202816 + IR-162016
2. If a full complement roller bearing is required, please contact NSK.



INNER RINGS FOR NEEDLE ROLLER BEARINGS

DRAWN CUP TYPE

			Pages
Metric Design (FIR)	Bore Diameter	7–45 mm	B130–B135
Inch Design (IR)	Bore Diameter	9.525–38.100 mm	B136–B137

SOLID TYPE

Metric Design (FIR)	Bore Diameter	5–220 mm	B138–B151
Metric Design (FIR)	Bore Diameter	8–50 mm	B152–B155
Inch Design (IR)	Bore Diameter	9.525–203.200 mm	B156–B161

For use when shafts cannot be finished with proper accuracy and hardness, NSK produces inner rings that are made from carefully selected bearing steel and finished with heat treatment and grinding work.

TYPE

The various types of inner rings for needle roller bearings are shown in **Table 1**.

ACCURACY

Dimensional accuracy of the inner rings of NSK needle roller bearings conforms with ISO (refer to pages A26 and A27). Tolerances for inch inner rings are shown in the dimensional table for drawn cup bearings, and in **Table 3** on page B92 for solid bearings.

Table 1 Types of Inner Ring

Code	Details
FIR	Metric needle bearing combination
IR	Inch needle bearing combination

MOUNTING

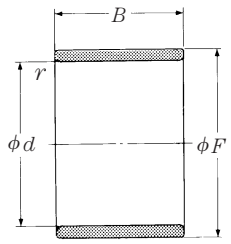
The inner ring is mounted with the unmarked side placed onto the shaft shoulder.

Fitting tolerance for the drawn cup needle bearing shaft should be h6 or js6 (j6). When requesting a tight fit for the inner ring, please consult NSK.

Each shaft fitting tolerance for the inner ring for solid needle bearings is shown in **Table 5.2** on page A33.

FIR

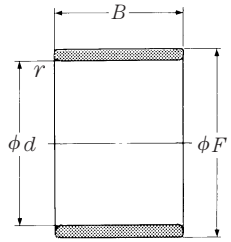
Bore Diameter 7 — 15 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (g)	Matching Drawn Cup Needle Roller Bearings		
d	F h5	B	r min		approx	F	FJ	FJL
7	10	10.5	0.3	**FIR-071010**	3.3	FH-1010	FJ-1010	FJL-1010
	10	12.5	0.3	**FIR-071012**	3.9	—	—	FJL-1012L
	10	12.5	0.3	**FIR-071015**	4.8	—	FJ-1015	FJL-1015
8	12	10.5	0.3	**FIR-081210**	5.1	—	FJ-1210	—
	12	12.5	0.3	**FIR-081212**	6.1	FJ-1212	FJH-1212	FJL-1212L
	12	15.5	0.3	**FIR-081215**	7.6	—	—	FJL-1215
10	13	12.5	0.3	**FIR-101312**	5.3	F-1312	FJ-1312	—
	14	12.5	0.3	**FIR-101412**	7.3	F-1412	FJ-1412	—
	14	13.5	0.3	**FIR-101413**	7.9	F-1413	—	—
	14	16.5	0.3	**FIR-101416**	9.7	—	—	FJL-1416L
	14	20.5	0.3	**FIR-101420**	12	—	—	FJL-1420L
	15	10.5	0.3	**FIR-101510**	8.0	—	—	FJL-1510L
	15	12.5	0.3	**FIR-101512**	9.5	F-1512	FJ-1512	FJL-1512L
	15	15.5	0.3	**FIR-101515**	12	—	—	FJL-1515
	15	20.5	0.3	**FIR-101520**	16	—	—	FJL-1520
	15	25.5	0.3	**FIR-101525**	19	—	—	FJL-1525L
12	15	12.5	0.3	**FIR-121512**	6.2	F-1512	FJ-1512	FJL-1512L
	15	16.5	0.3	**FIR-121516**	8.2	F-1516	FJ-1516	—
	16	12.5	0.3	**FIR-121612**	8.6	F-1612	FJ-1612	—
	16	16.5	0.3	**FIR-121616**	11	F-1616	FJ-1616	FJL-1616L
	16	20.5	0.3	**FIR-121620**	14	—	—	FJL-1620L
	17	15.5	0.3	**FIR-121715**	14	—	—	FJL-1715
	17	20.5	0.3	**FIR-121720**	18	—	—	FJL-1720L
	17	25.5	0.3	**FIR-121725**	23	—	—	FJL-1725L
15	18	12.5	0.3	**FIR-151812**	7.6	F-1812	FJ-1812	—
	18	15.5	0.3	**FIR-151815**	9.4	—	—	FJL-1815
	18	16.5	0.3	**FIR-151816**	10	F-1816	FJ-1816	—
	18	20.5	0.3	**FIR-151820**	12	—	—	FJL-1820L
	18	25.5	0.3	**FIR-151825**	15	—	—	FJL-1825
	20	15.5	0.3	**FIR-152015**	17	—	—	FJL-2015
	20	20.5	0.3	**FIR-152020**	22	F-2020	FJ-2020	FJL-2020
	20	25.5	0.3	**FIR-152025**	27	—	—	FJL-2025L
	20	30.5	0.3	**FIR-152030**	33	—	—	FJL-2030L

FIR

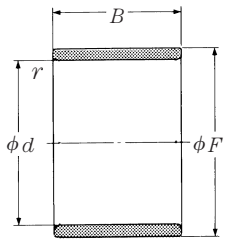
Bore Diameter 17 — 30 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (g)	Matching Drawn Cup Needle Roller Bearings		
d	F h5	B	r min		approx	F	FJ	FJL
17	20	12.5	0.3	**FIR-172012**	8.5	F-2012	FJ-2012	—
	20	16.5	0.3	**FIR-172016**	11	F-2016	FJ-2016	—
	20	20.5	0.3	**FIR-172020**	14	F-2020	FJ-2020	FJL-2020
	22	10.5	0.3	**FIR-172210**	12	F-2210	—	FJL-2210L
	22	12.5	0.3	**FIR-172212**	15	F-2212	FJ-2212	—
	22	15.5	0.3	**FIR-172215**	18	—	—	FJL-2215L
	22	16.5	0.3	**FIR-172216**	20	F-2216	FJ-2216	—
	22	20.5	0.3	**FIR-172220**	24	F-2220	FJ-2220	FJL-2220L
	22	25.5	0.3	**FIR-172225**	30	—	—	FJL-2225L
	22	30.5	0.3	**FIR-172230**	36	—	—	FJL-2230L
20	25	10.5	0.3	**FIR-202510**	14	—	—	FJL-2510
	25	15.5	0.3	**FIR-202515**	21	—	—	FJL-2515L
	25	16.5	0.3	**FIR-202516**	23	F-2516	FJ-2516	—
	25	20.5	0.3	**FIR-202520**	28	F-2520	FJ-2520	FJL-2520
	25	25.5	0.3	**FIR-202525**	35	—	—	FJL-2525L
	25	26.5	0.3	**FIR-202526**	36	F-2526	FJ-2526	—
	25	30.5	0.3	**FIR-202530**	42	—	—	FJL-2530
22	28	16.5	0.3	**FIR-222816**	30	F-2816	FJ-2816	—
	28	20.5	0.3	**FIR-222820**	38	F-2820	FJ-2820	FJL-2820L
	28	26.5	0.3	**FIR-222826**	49	F-2826	FJ-2826	—
	28	30.5	0.3	**FIR-222830**	56	—	—	FJL-2830L
25	30	15.5	0.3	**FIR-253015**	26	—	—	FJL-3015L
	30	20.5	0.3	**FIR-253020**	34	F-3020	FJ-3020	FJL-3020L
	30	25.5	0.3	**FIR-253026**	43	—	—	FJL-3025L
	30	26.5	0.3	**FIR-253026**	44	F-3026	FJ-3026	—
	30	30.5	0.3	**FIR-253030**	51	—	—	FJL-3030L
28	32	20.5	0.3	**FIR-283220**	30	—	—	FJL-3220
	32	30.5	0.3	**FIR-283230**	45	—	—	FJL-3230
30	35	12.5	0.6	**FIR-303512**	25	F-3512	FJ-3512	FJL-3512
	35	15.5	0.6	**FIR-303515**	31	—	—	FJL-3515
	35	20.5	0.6	**FIR-303520**	41	F-3520	FJ-3520	FJL-3520L
	35	25.5	0.6	**FIR-303525**	51	—	—	FJL-3525L
	35	26.5	0.6	**FIR-303526**	53	F-3526	FJ-3526	—
	35	30.5	0.6	**FIR-303530**	61	—	—	FJL-3530L

FIR

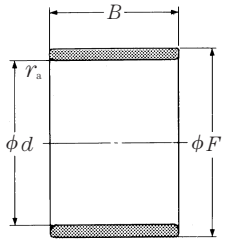
Bore Diameter 35 — 45 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (g)	Matching Drawn Cup Needle Roller Bearings		
d	F h5	B	r min		approx	F	FJ	FJL
35	40	15.5	0.6	**FIR-354015**	36	—	—	FJL-4015L
	40	20.5	0.6	**FIR-354020**	47	F-4020	FJ-4020	FJL-4020L
	40	25.5	0.6	**FIR-354025**	58	—	—	FJL-4025L
	40	26.5	0.6	**FIR-354026**	61	F-4026	FJ-4026	—
	40	30.5	0.6	**FIR-354030**	70	—	—	FJL-4030L
40	45	20.5	0.6	**FIR-404520**	53	F-4520	FJ-4520	FJL-4520L
	45	25.5	0.6	**FIR-404525**	66	—	—	FJL-4525L
	45	26.5	0.6	**FIR-404526**	67	F-4526	FJ-4526	—
	45	30.5	0.6	**FIR-404530**	79	—	—	FJL-4530L
45	50	12.5	0.6	**FIR-455012**	36	—	—	FJL-5012L
	50	15.5	0.6	**FIR-455015**	45	—	—	FJL-5015L
	50	20.5	0.6	**FIR-455020**	59	—	FJ-5020	FJL-5020L
	50	25.5	0.6	**FIR-455025**	74	—	—	FJL-5025L
	50	30.5	0.6	**FIR-455030**	89	—	—	FJL-5030L

IR

Bore Diameter 9.525 — 38.100 mm



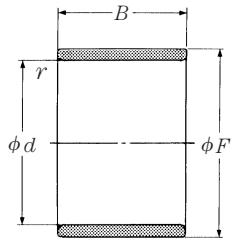
Boundary Dimensions (mm, inch) d $^{0}_{-0.013}$		F $^{0}_{-0.013}$		B $^{0}_{-0.25}$	Inner Ring Numbers	Mass (g) approx	Fillet Radius of Shaft (mm) r_a max
9.525	0.3750	14.288	0.5625	13.08	**IR-68**	9.1	0.6
		15.875	0.6250	13.08	**IR-68-1**	13	0.6
		14.288	0.5625	19.43	**IR-612**	13	0.6
		15.875	0.6250	19.43	**IR-612-1**	19	0.6
12.700	0.5000	19.050	0.7500	13.08	**IR-88**	16	1.0
		19.050	0.7500	19.43	**IR-812**	24	1.0
15.875	0.6250	22.225	0.8750	19.43	**IR-1012**	29	1.0
		22.225	0.8750	25.78	**IR-1016**	38	1.0
19.050	0.7500	25.400	1.0000	13.08	**IR-128**	23	1.0
		25.400	1.0000	19.43	**IR-1212**	34	1.0
		25.400	1.0000	25.78	**IR-1216**	44	1.0
		25.400	1.0000	32.13	**IR-1220**	55	1.0
		25.400	1.0000	38.48	**IR-1224**	66	1.0
20.638	0.8125	25.400	1.0000	19.43	**IR-1312**	26	1.0
		25.400	1.0000	25.78	**IR-1316**	35	1.0
22.225	0.8750	28.575	1.1250	25.78	**IR-1416**	51	1.0
23.812	0.9375	28.575	1.1250	25.78	**IR-1516**	39	1.0
25.400	1.0000	31.750	1.1250	19.42	**IR-1612**	43	1.0
		31.750	1.1250	25.78	**IR-1616**	57	1.0
28.575	1.1250	34.925	1.3750	19.43	**IR-1812**	48	1.0
		34.925	1.3750	25.78	**IR-1816**	64	1.0
		34.925	1.3750	32.13	**IR-1820**	79	1.0
30.162	1.1875	38.100	1.5000	25.78	**IR-1916**	85	1.0
		38.100	1.5000	32.13	**IR-1920**	105	1.0
31.750	1.2500	38.100	1.5000	25.78	**IR-2016**	70	1.5
		38.100	1.5000	32.13	**IR-2020**	87	1.5
34.925	1.3750	41.275	1.6250	32.13	**IR-2220**	95	1.5
36.512	1.4375	44.450	1.7500	25.78	**IR-2316**	100	1.5
		44.450	1.7500	38.48	**IR-2324**	150	1.5
38.100	1.5000	44.450	1.7500	25.78	**IR-2416**	83	1.5
		44.450	1.7500	38.48	**IR-2424**	125	1.5

Matching Drawn Cup Needle Roller Bearings			
B	BH	J	JH
B-98	BH-98	J-98	JH-98
B-108	BH-108	J-108	—
B-912	BH-912	—	—
B-1012	BH-1012	J-1012	—
B-128	—	J-128	—
B-1212	—	J-1212	JH-1212
B-1412	BH-1412	J-1412	JH-1412
B-1416	—	J-1416	JH-1416
B-168	BH-168	—	—
B-1612	BH-1612	J-1612	JH-1612
B-1616	BH-1616	J-1616	JH-1616
—	BH-1620	—	—
—	BH-1624	—	—
B-1612	BH-1612	J-1612	JH-1612
B-1616	BH-1616	J-1616	JH-1616
B-1816	BH-1816	J-1816	JH-1816
B-1816	BH-1816	J-1816	JH-1816
B-2012	BH-2012	J-2012	—
B-2016	BH-2016	J-2016	JH-2016
B-2212	BH-2212	—	JH-2212
B-2216	BH-2216	—	JH-2216
B-2220	BH-2220	—	—
B-2416	—	J-2416	—
B-2420	—	J-2420	—
B-2416	—	J-2416	—
B-2420	—	J-2420	—
B-2620	—	—	—
B-2816	—	J-2816	—
B-2824	—	J-2824	—
B-2816	—	J-2816	—
B-2824	—	J-2824	—

Remarks Inner rings can be used with Y and YH types (pages B76~B79).

FIR

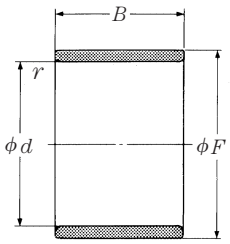
Bore Diameter 5 — 15 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (kg)	Matching Solid Needle Roller Bearings			
d	F	B	r min		approx	RLM	RNA	RNAF	RNAFW
5	8	12	0.15	**FIR-5812-1**	0.003	RLM81512-1			
	8	16	0.15	**FIR-5816-1**	0.004	RLM81516-1			
7	10	12	0.15	**FIR-71012-1**	0.004	RLM101712-1			
	10	16	0.15	**FIR-71016-1**	0.005	RLM101716-1			
9	12	16	0.3	**FIR-91216-1**	0.006	RLM121916-1			
10	14	12	0.3	**FIR-101412-1**	0.007			RNAF142612	
	14	13	0.3	**FIR-101413-1**	0.008		RNA4900	RNAF142213	
	14	14	0.3	**FIR-101414-1**	0.008		RNA4900TT		
	14	16	0.3	**FIR-101416-1**	0.009		RNA5900		
	14	20	0.3	**FIR-101420-1**	0.012				RNAFW142220
	14	22	0.3	**FIR-101422-1**	0.013		RNA6900		
12	16	12	0.3	**FIR-121612-1**	0.008			RNAF162812	
	16	13	0.3	**FIR-121613-1**	0.009		RNA4901	RNAF162413	
	16	14	0.3	**FIR-121614-1**	0.010		RNA4901TT		
	16	16	0.3	**FIR-121616-1**	0.011		RNA5901		
	16	20	0.3	**FIR-121620-1**	0.014				RNAFW162420
	16	22	0.3	**FIR-121622-1**	0.015		RNA6901		
	16	23	0.3	**FIR-121623-1**	0.016		RNA6901TT		
15	20	12	0.3	**FIR-152012-1**	0.013			RNAF203212	
	20	13	0.3	**FIR-152013-1**	0.014		RNA4902	RNAF202813	
	20	14	0.3	**FIR-152014-1**	0.015		RNA4902TT		
	20	18	0.3	**FIR-152018-1**	0.019		RNA5902		
	20	23	0.3	**FIR-152023-1**	0.025		RNA5902		
	20	24	0.3	**FIR-152024-1**	0.026		RNA6902TT		
	20	26	0.3	**FIR-152026-1**	0.028				RNAFW202826

FIR

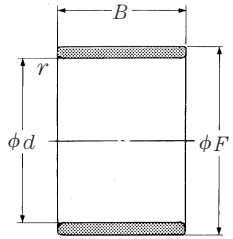
Bore Diameter 17 — 25 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (kg)	Matching Solid Needle Roller Bearings			
d	F	B	r min		approx	RLM	RNA	RNAF	RNAFW
17	22	13	0.3	**FIR-172213-1**	0.016		RNA4903	RNAF223013	
	22	14	0.3	**FIR-172214-1**	0.017		RNA4903TT		
	22	16	0.3	**FIR-172216-1**	0.019			RNAF223516	
	22	18	0.3	**FIR-172218-1**	0.022		RNA5903		
	22	23	0.3	**FIR-172223-1**	0.027		RNA6903		
	22	24	0.3	**FIR-172224-1**	0.029		RNA6903TT		
	22	26	0.3	**FIR-172226-1**	0.031				RNAFW223026
	22	32	0.3	**FIR-172232-1**	0.038				RNAFW223532
20	25	16	0.3	**FIR-202516-1**	0.022			RNAF253716	
	25	17	0.3	**FIR-202517-1**	0.023		RNA4904	RNAF253517	
	25	18	0.3	**FIR-202518-1**	0.025		RNA4904TT		
	25	23	0.3	**FIR-202523-1**	0.032		RNA5904		
	25	26	0.3	**FIR-202526-1**	0.036				RNAFW253526
	25	30	0.3	**FIR-202530-1**	0.041		RNA6904		
	25	31	0.3	**FIR-202531-1**	0.043		RNA6904TT		
	25	32	0.3	**FIR-202532-1**	0.044				RNAFW253732
22	28	17	0.3	**FIR-222817-1**	0.031		RNA49/22		
	28	18	0.3	**FIR-222818-1**	0.033		RNA49/22TT		
	28	23	0.3	**FIR-222823-1**	0.042		RNA59/22		
	28	30	0.3	**FIR-222830-1**	0.055		RNA69/22		
	28	31	0.3	**FIR-222831-1**	0.057		RNA69/22TT		
25	29	20	0.3	**FIR-252920-1**	0.026	RLM293820-1			
	30	16	0.3	**FIR-253016-1**	0.027			RNAF304216	
	30	17	0.3	**FIR-253017-1**	0.029		RNA4905	RNAF304017	
	30	18	0.3	**FIR-253018-1**	0.030		RNA4905TT		
	30	23	0.3	**FIR-253023-1**	0.039		RNA5905		
	30	26	0.3	**FIR-253026-1**	0.044				RNAFW304026
	30	30	0.3	**FIR-253030-1**	0.051		RNA6905		
	30	31	0.3	**FIR-253031-1**	0.052		RNA6905TT		
	30	32	0.3	**FIR-253032-1**	0.054				RNAFW304232

FIR

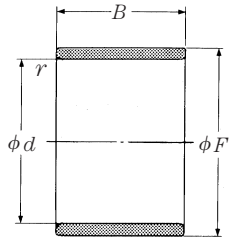
Bore Diameter 28 — 38 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (kg)	Matching Solid Needle Roller Bearings			
d	F	B	r min		approx	RLM	RNA	RNAF	RNAFW
28	32	17	0.3	**FIR-283217-1**	0.025		RNA49/28		
	32	18	0.3	**FIR-283218-1**	0.026		RNA49/28TT		
	32	23	0.3	**FIR-283223-1**	0.034		RNA59/28		
	32	30	0.3	**FIR-283230-1**	0.044		RNA69/28		
	32	31	0.3	**FIR-283231-1**	0.046		RNA69/28TT		
30	35	13	0.3	**FIR-303513-1**	0.026			RNAF354513	
	35	16	0.3	**FIR-303516-1**	0.032			RNAF354716	
	35	17	0.3	**FIR-303517-1**	0.034		RNA4906	RNAF354517	
	35	18	0.3	**FIR-303518-1**	0.036		RNA4906TT		
	35	23	0.3	**FIR-303523-1**	0.046		RNA5906		
	35	26	0.3	**FIR-303526-1**	0.052				RNAFW354526
	35	30	0.3	**FIR-303530-1**	0.060		RNA6906		
	35	31	0.3	**FIR-303531-1**	0.062		RNA6906TT		
	35	32	0.3	**FIR-303532-1**	0.064				RNAFW354732
32	37	30	0.3	**FIR-323730-1**	0.063	RLM374730-1			
	40	20	0.6	**FIR-324020-1**	0.071		RNA49/32		
	40	21	0.6	**FIR-324021-1**	0.074		RNA49/32TT		
	40	27	0.6	**FIR-324027-1**	0.095		RNA59/32		
	40	36	0.6	**FIR-324036-1**	0.127		RNA69/32		
	40	37	0.6	**FIR-324037-1**	0.131		RNA69/32TT		
35	40	17	0.3	**FIR-354017-1**	0.039			RNAF405017	
	40	20	0.3	**FIR-354020-1**	0.046			RNAF405520	
	40	34	0.3	**FIR-354034-1**	0.078				RNAFW405034
	40	40	0.3	**FIR-354040-1**	0.092				RNAFW405540
	42	20	0.6	**FIR-354220-1**	0.066		RNA4907		
	42	21	0.6	**FIR-354221-1**	0.069		RNA4907TT		
	42	27	0.6	**FIR-354227-1**	0.089		RNA5907		
	42	36	0.6	**FIR-354236-1**	0.119		RNA6907		
	42	37	0.6	**FIR-354237-1**	0.122		RNA6907TT		
38	43	20	0.3	**FIR-384320-1**	0.050	RLM435320-1			
	43	30	0.3	**FIR-384330-1**	0.074	RLM435330-1			

FIR

Bore Diameter 40 — 50 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (kg)	Matching Solid Needle Roller Bearings			
d	F	B	r min		approx	RLM	RNA	RNAF	RNAFW
40	45	17	0.3	**FIR-404517-1**	0.044			RNAF455517	
	45	20	0.3	**FIR-404520-1**	0.052			RNAF456220	
	45	34	0.3	**FIR-404534-1**	0.089				RNAFW455534
	45	40	0.3	**FIR-404540-1**	0.104				RNAFW456240
	48	22	0.6	**FIR-404822-1**	0.095		RNA4908		
	48	23	0.6	**FIR-404823-1**	0.099		RNA4908TT		
	48	30	0.6	**FIR-404830-1**	0.129		RNA5908		
	48	40	0.6	**FIR-404840-1**	0.173		RNA6908		
	48	41	0.6	**FIR-404841-1**	0.177		RNA6908TT		
42	47	20	0.3	**FIR-424720-1**	0.055	RLM475720-1			
	47	30	0.3	**FIR-424730-1**	0.082	RLM475730-1			
45	50	20	0.3	**FIR-455020-1**	0.058			RNAF506220	
	50	25	0.3	**FIR-455025-1**	0.073	RLM506225			
	50	35	0.6	**FIR-455035-1**	0.102	RLM506235-1			
	50	40	0.3	**FIR-455040-1**	0.116				RNAFW506240
	52	22	0.6	**FIR-455222-1**	0.092		RNA4909		
	52	23	0.6	**FIR-455223-1**	0.096		RNA4909TT		
	52	30	0.6	**FIR-455230-1**	0.125		RNA5909		
	52	40	0.6	**FIR-455240-1**	0.166		RNA6909		
	52	41	0.6	**FIR-455241-1**	0.171		RNA6909TT		
	55	40	0.6	**FIR-455540-1**	0.245				RNAFW557240
50	55	20	0.6	**FIR-505520-1**	0.064			RNAF556820	
	55	25	0.5	**FIR-505525-1**	0.080	RLM556825-1			
	55	35	0.3	**FIR-505535-1**	0.113	RLM556835-1			
	55	40	0.6	**FIR-505540-1**	0.129				RNAFW556840
	58	22	0.6	**FIR-505822-1**	0.116		RNA4910		
	58	23	0.6	**FIR-505823-1**	0.122		RNA4910TT		
	58	30	0.6	**FIR-505830-1**	0.159		RNA5910		
	58	40	0.6	**FIR-505840-1**	0.212		RNA6910		
	58	41	0.6	**FIR-505841-1**	0.217		RNA6910TT		
	60	20	1	**FIR-506020-1**	0.135			RNAF607820	
	60	40	1	**FIR-506040-1**	0.270				RNAFW607840